Basics Of DRAM Interfaces

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Outline

- Introduction
- DRAM Background
- DRAM Families
  - Main Memory DRAM
  - Graphics DRAM
  - LPDDR
  - TSV Solutions (HBM, HMC, Wide I/O)
- Summary and Conclusion

Introduction

- The DRAM world of 2015 is fragmented
- Important metrics
  - Memory density/modules => DDRn
  - Bandwidth => GDDR, HBM, HMC
  - Power => LPDDRn, Wide I/O
  - Integration / TSV => Wide I/O, HBM, HMC
- Widely used acronyms for memory families
  - **DDR**: Double-data-rate DRAM
  - **GDDR**: Graphics-DDR
  - **LPDDR**: Low-power DDR
  - **Wide I/O**: Wide I/O interface DRAM
  - **HBM**: High-bandwidth memory
  - **HMC**: Hybrid Memory Cube

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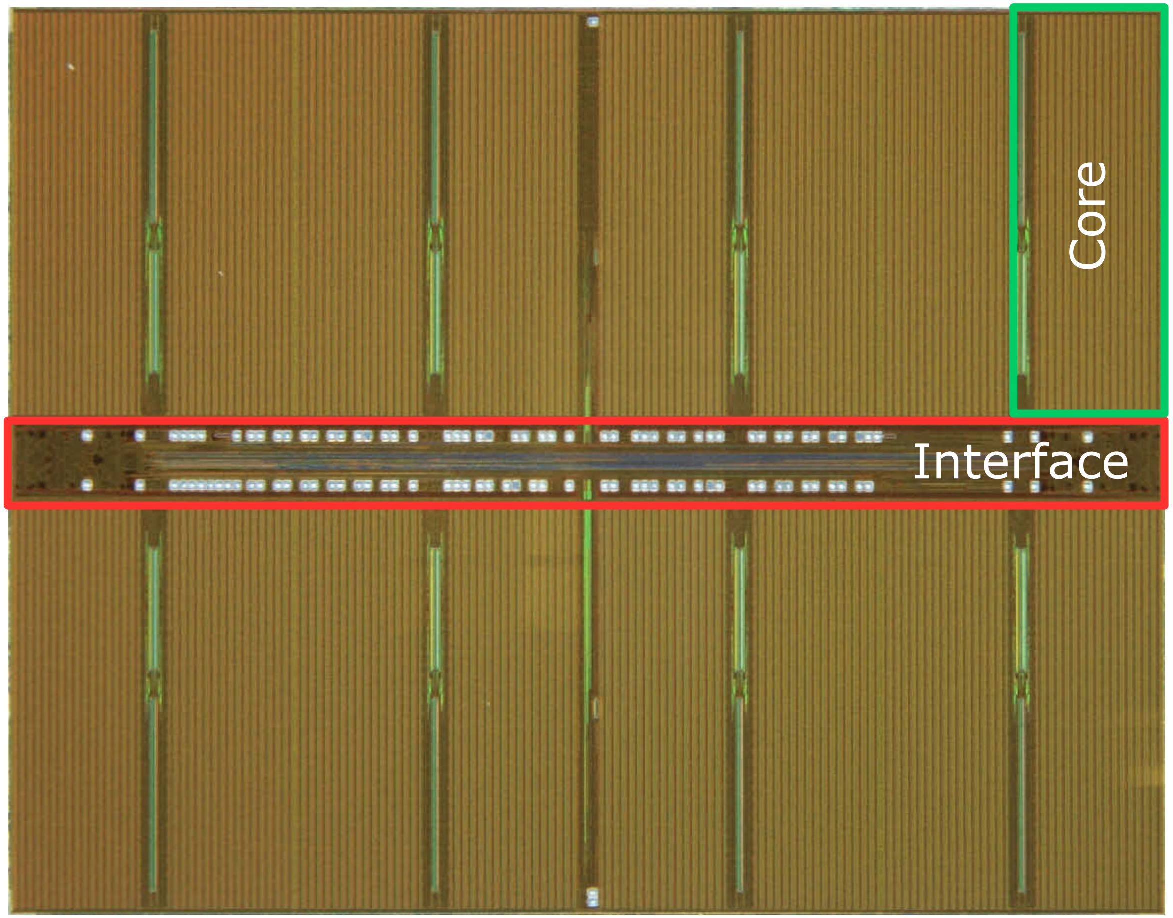
- DRAM Background
- DRAM Families
  - Main Memory DRAM
  - Graphics DRAM
  - LPDDR
  - TSV Solutions (HBM, HMC, Wide I/O)
- Summary and Conclusion

DRAM Background

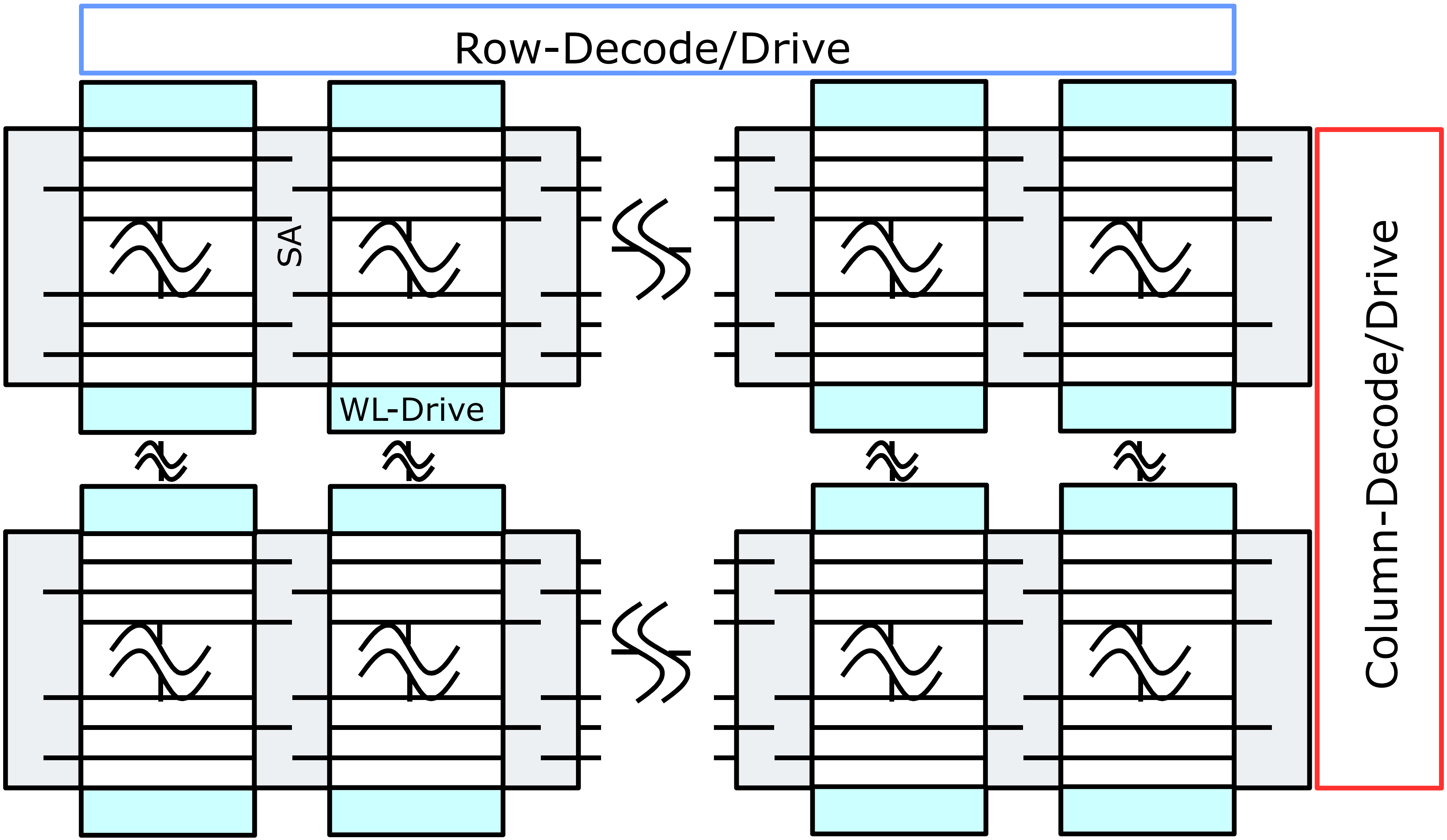
- DRAM core
- Generic DRAM commands
- Hiding the core from the interface
- Address/command processing

Typical DRAM Die

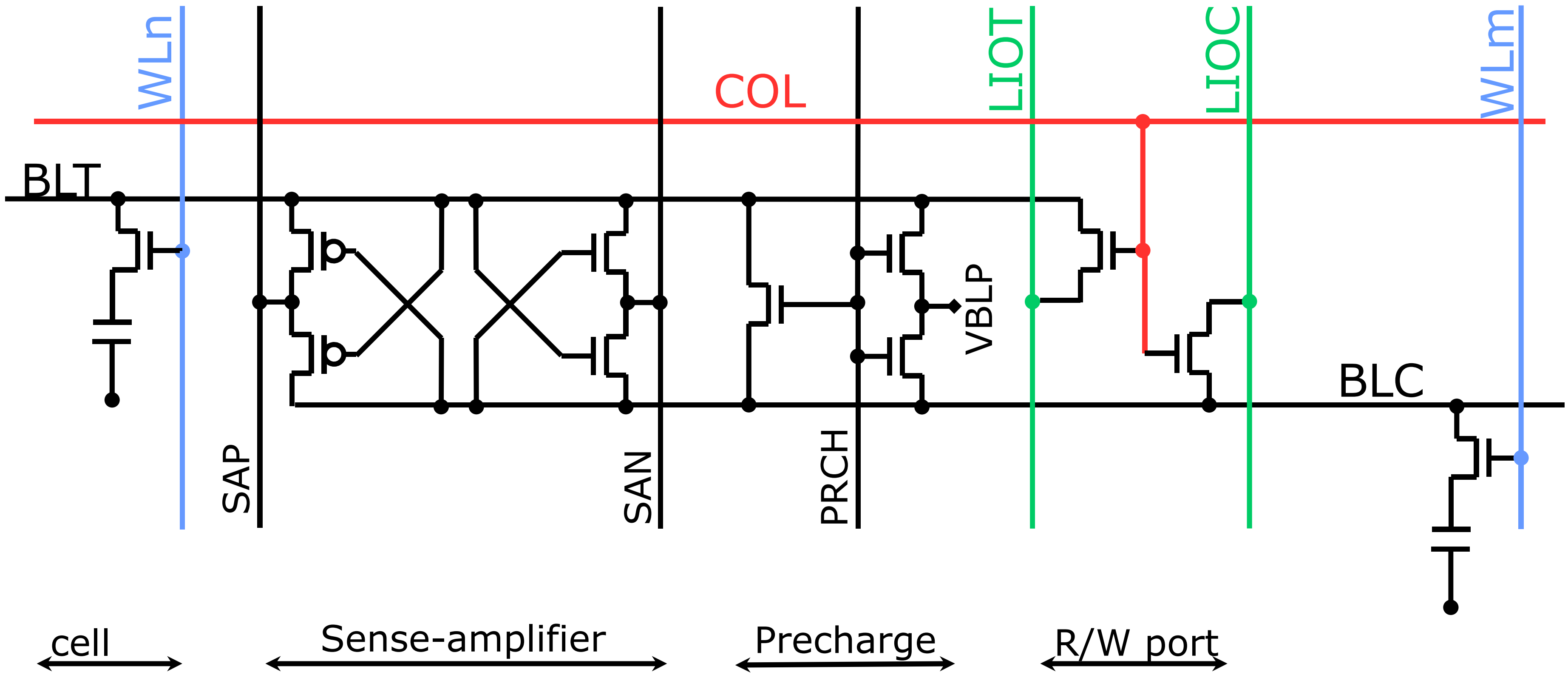


Example: 2Gb DDR3

Core (1)

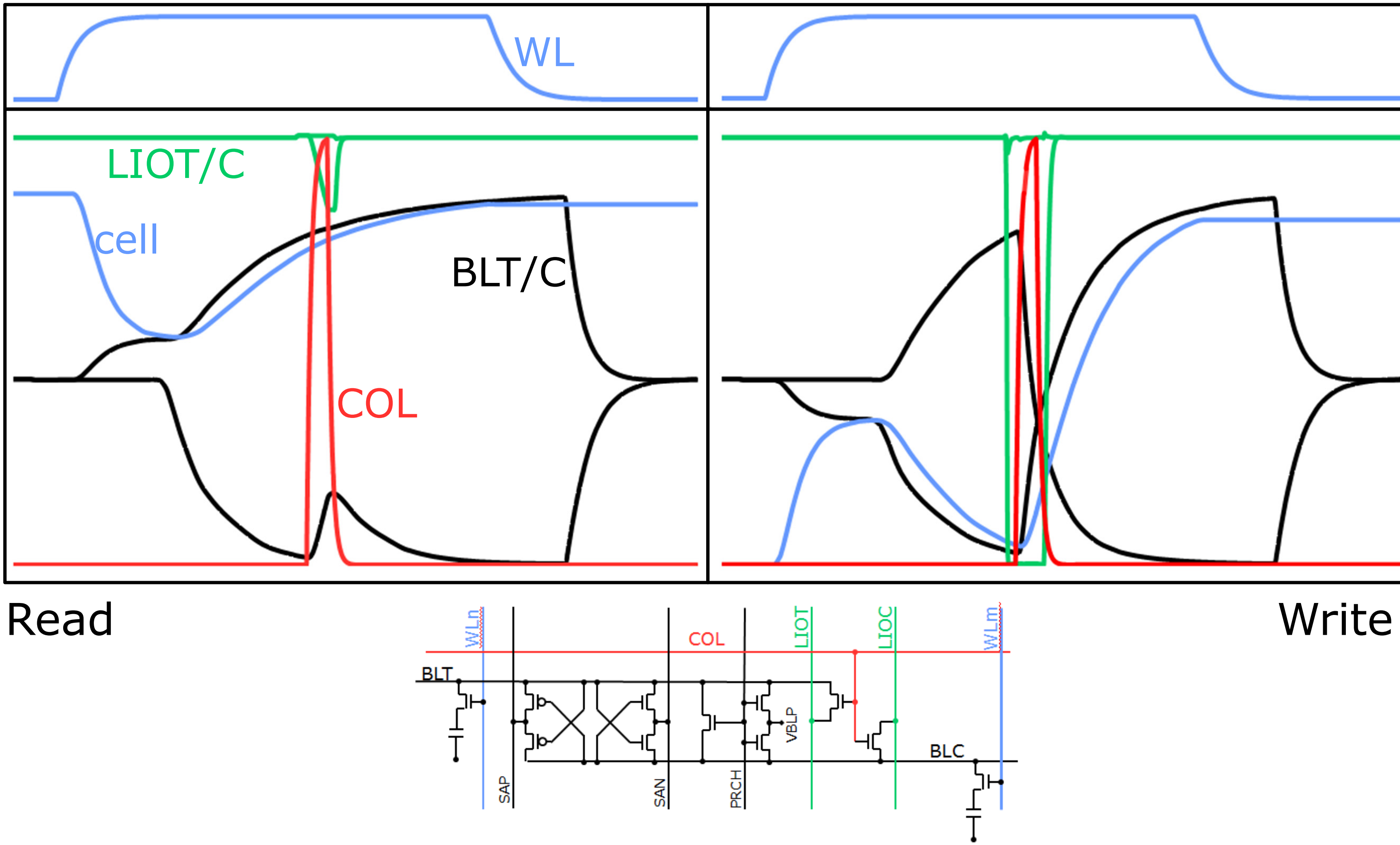


Core (2)



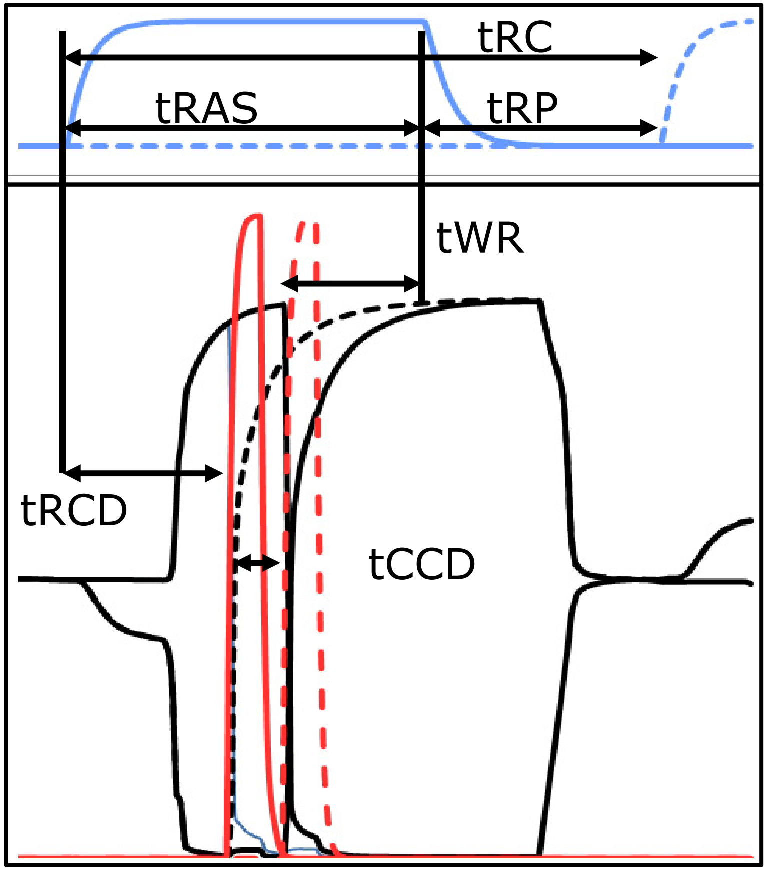
Typical today:
512b to 1024b/BL
1024 b/WL

Core (3)


WL
LIOT/C
cell
BLT/C
COL
Read
Write
WLn
COL
LIOT
LIOC
WLm
BLT
BLC
SAP
SAN
PRCH
VBLP

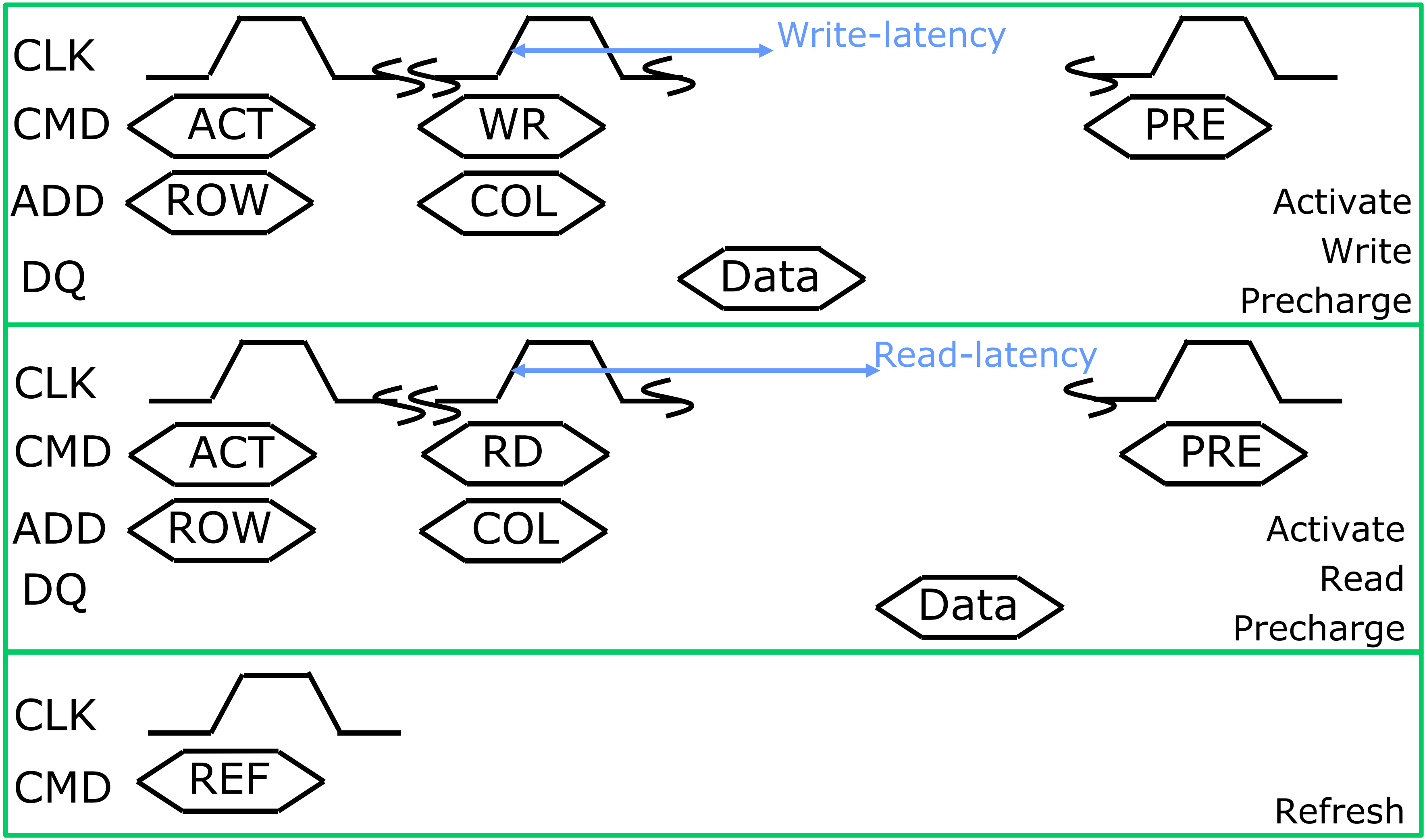
Core (4)

tRC:	Row cycle time	50ns
tRAS:	Row active time	30ns
tRP:	Row precharge time	20ns
tRCD:	Row/column delay	20ns
tCCD:	Column cycle time	5ns
tWR:	Write-back time	20ns



- Absolute numbers
  - Dependent on family
  - Defined at the external interface of the chip

Basic DRAM Command Set



Truth Table (1)

CS	RAS	CAS	WE	
H	H	H	H	DESELECT
L	H	H	H	NO OPERATION
L	L	H	H	ACTIVATE
L	L	H	L	PRECHARGE
L	H	L	H	READ
L	H	L	L	WRITE
L	L	L	H	REFRESH

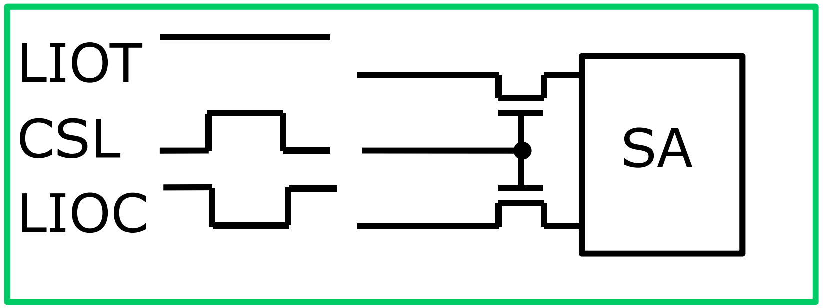
- Sub-set for basic commands
- Details vary between families
  - Example follows main memory style

Truth Table (2)

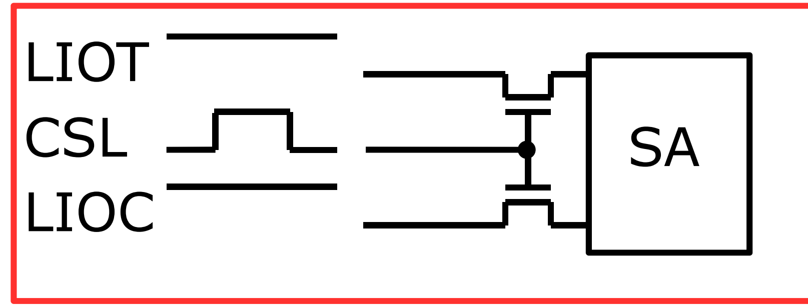
- Use free addresses to decode special functions
  - Fewer addresses needed during column commands
  - Example: Precharge automatically after R/W-access
- Self refresh
  - Refresh is performed automatically
- Programming of on-chip registers
  - Program timings, functionalities, …
- Write-masking
  - Block subset of data from being written to core

Write Masking

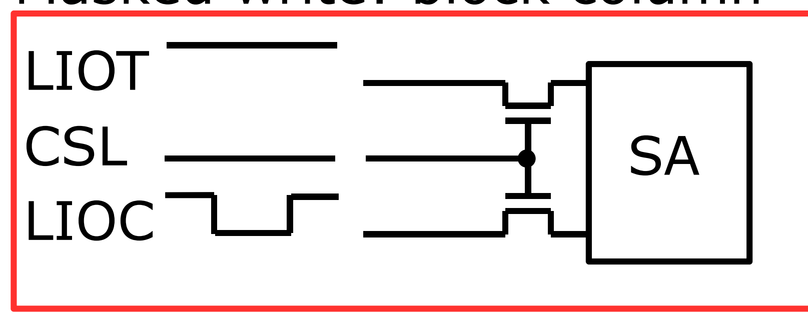
Normal write



Masked write: block data

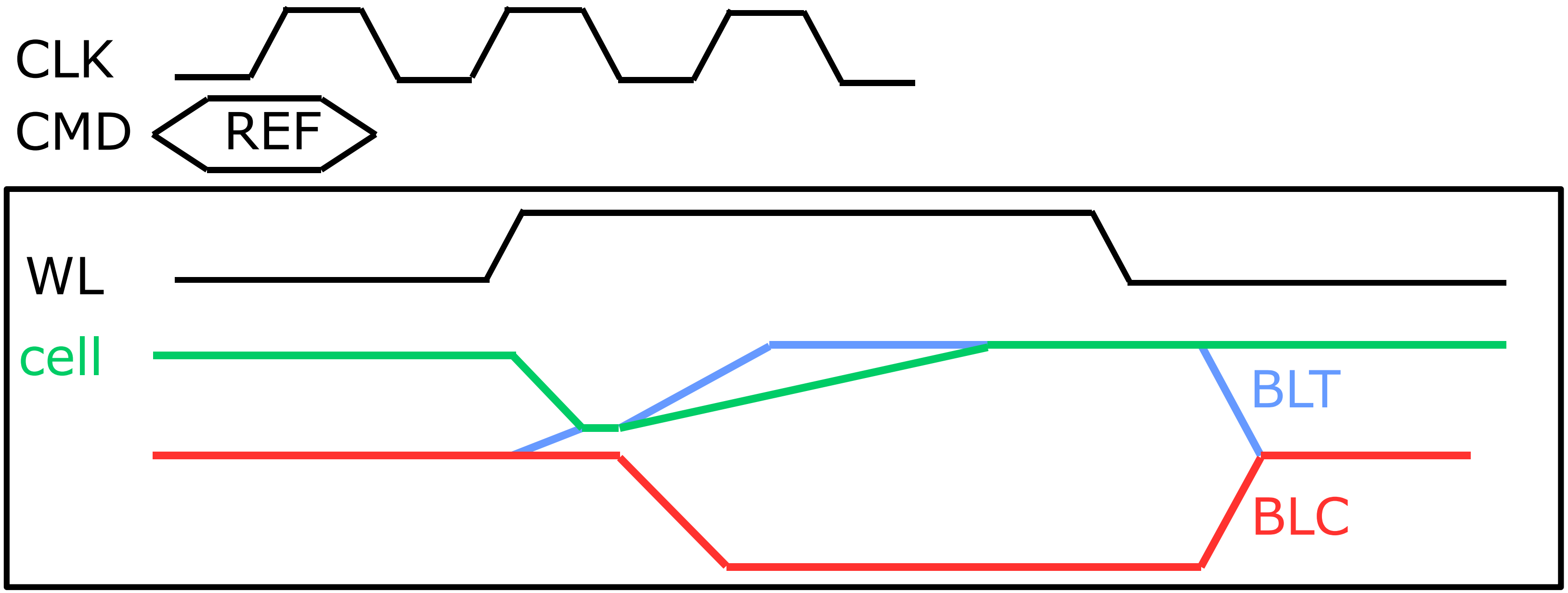


Masked write: block column



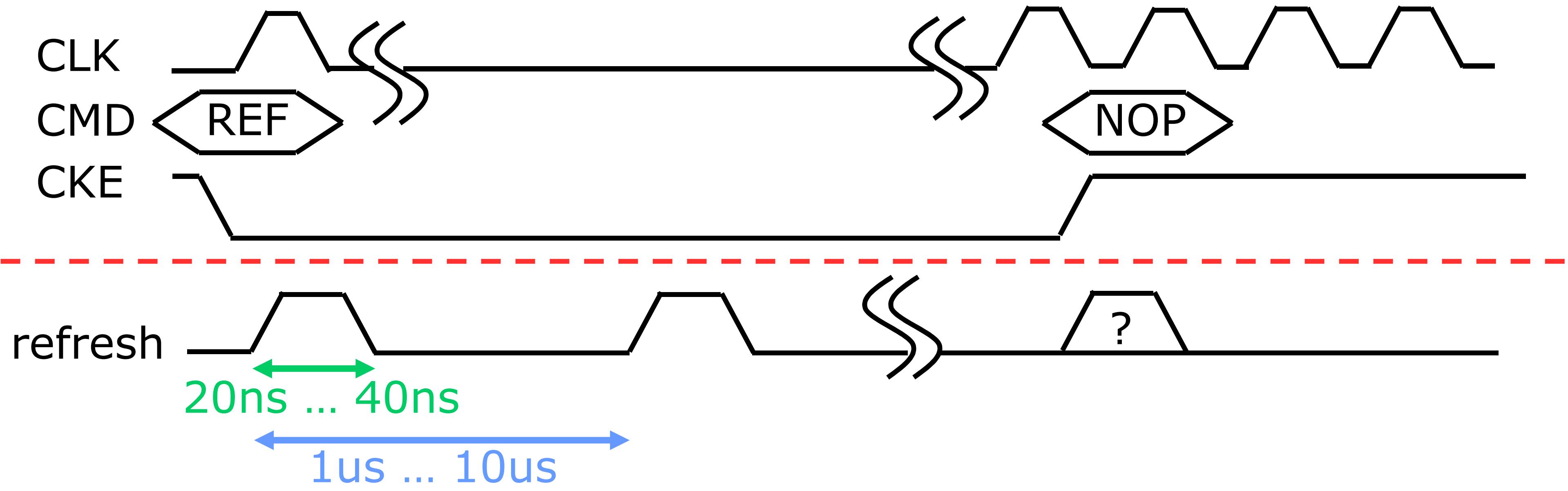
- Only a subset of the data is written to DRAM cell
- Other subset needs to be blocked from being written

Refresh



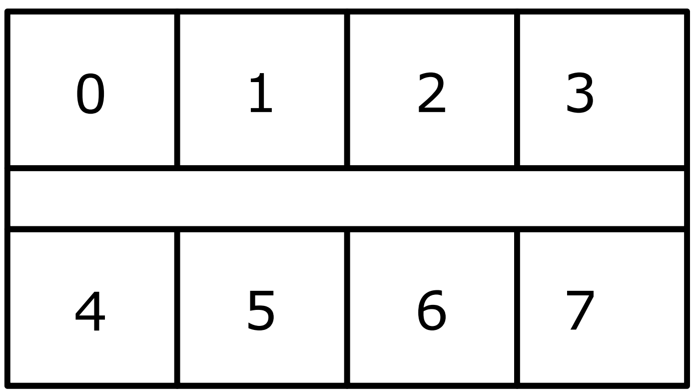
- DRAM cell capacitor requires periodic refresh
  - Typical refresh period is 64ms/cell
- All cells of a WL are refreshed in parallel
  - Refresh period of 8us for 8k row-addresses
- Refresh address is taken from counter to ensure periodicity
- WL is self-timed by (typically) asynchronous timer

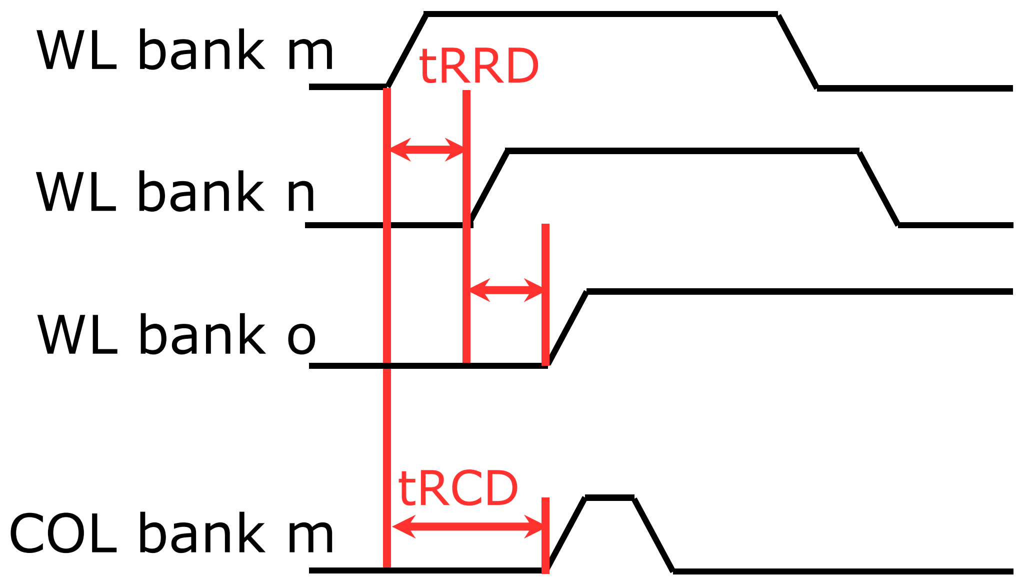
Self Refresh



- DRAM maintains information
  - Periodic refresh performed automatically until SR-exit
  - Commands other than SR-exit are not allowed
- Internal refresh activity is asynchronous to commands
  - Internal refresh may be on-going during SR-exit
- State with lowest current consumption

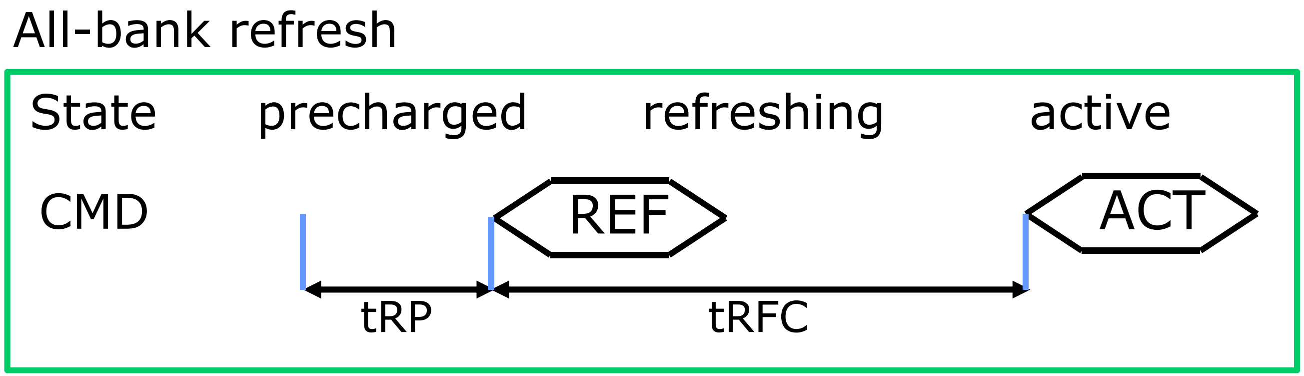
Hiding the Core: Banks (1)





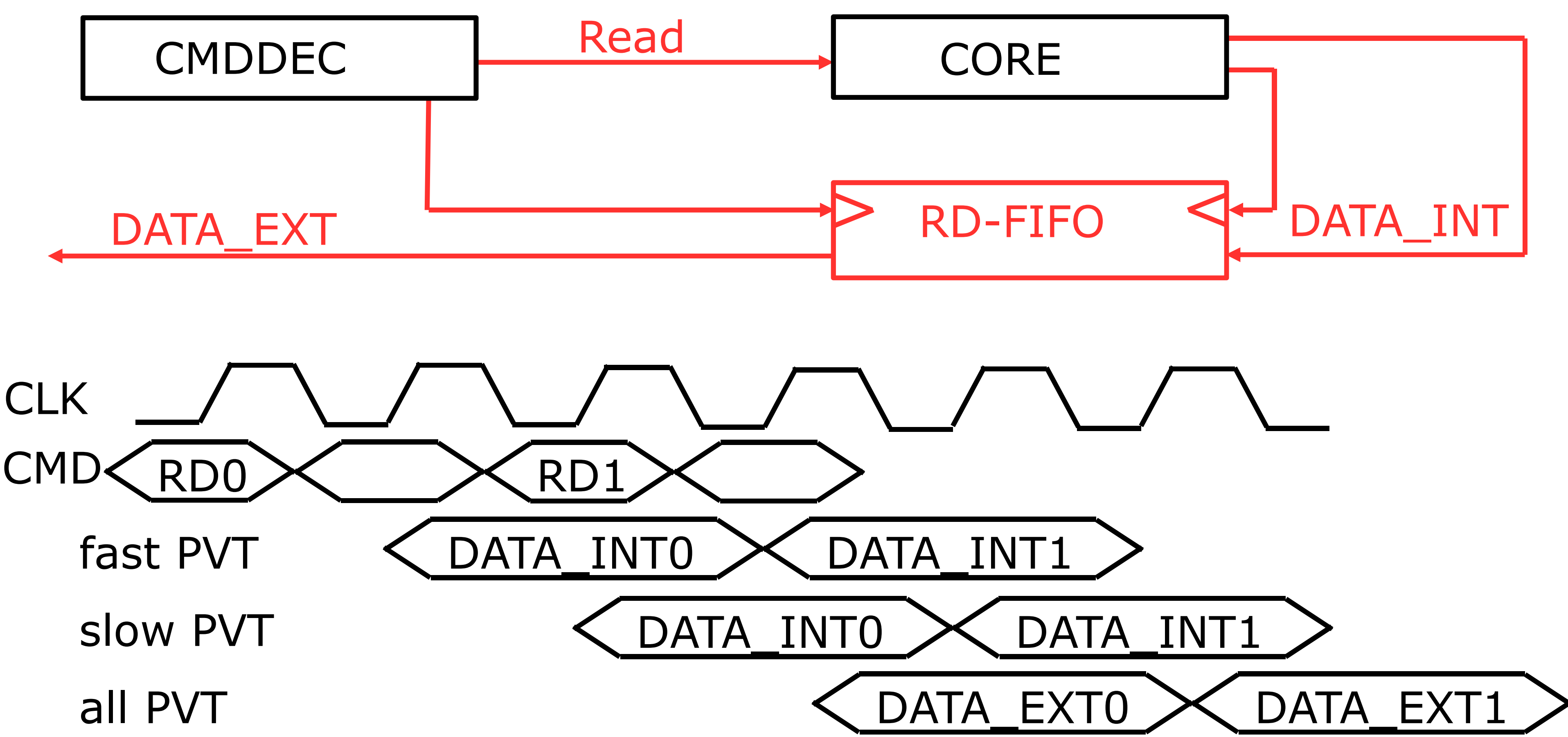
- Memory is partitioned into banks with independent rows
- Rows in different banks can be rapidly activated
  - tRRD ~ 10% of tRC
- Activate/precharge and column access may overlap

Hiding the Core: Banks (2)



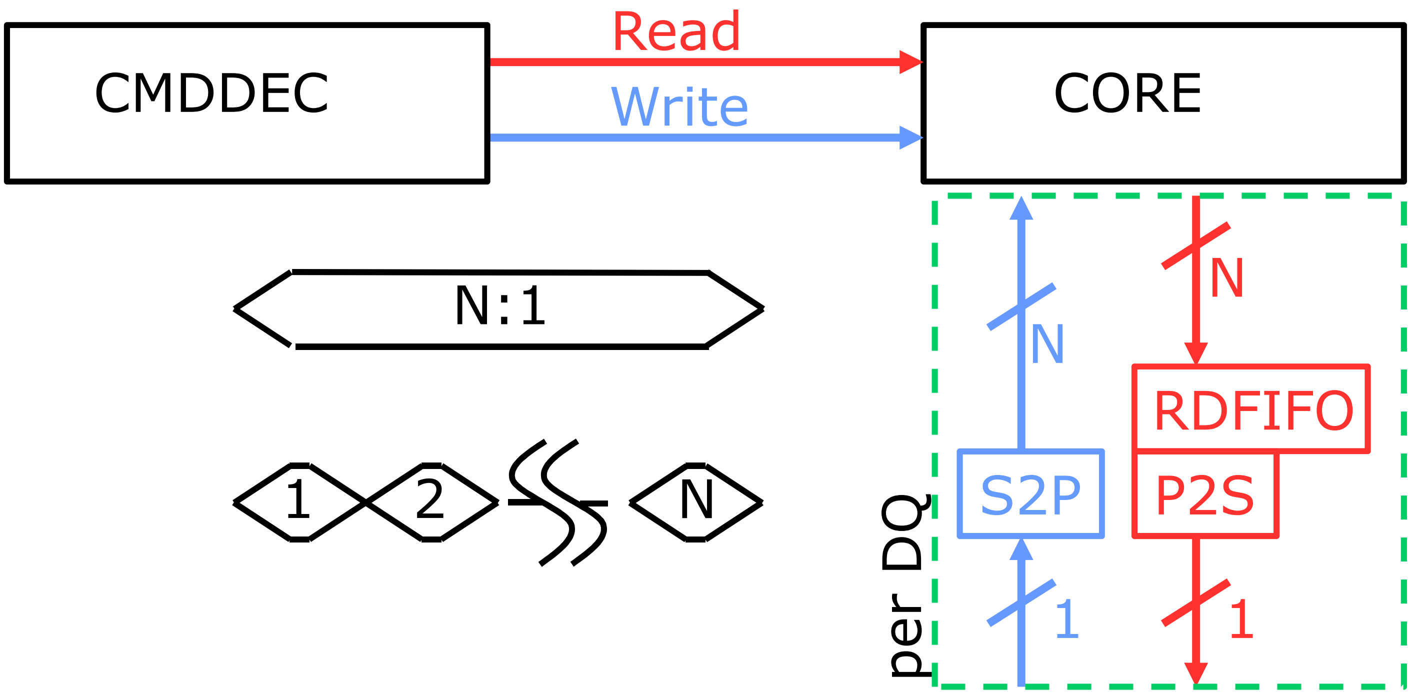
- All-bank refresh is the conventional implementation
  - All banks need to be closed before refresh
  - Simple protocol for controller, but loss of performance
- Per-bank refresh is an alternative technique
  - Only refreshes addressed bank
  - Other banks may remain open
  - Enables concurrent refresh and read/write to other banks

Hiding the Core: PVT



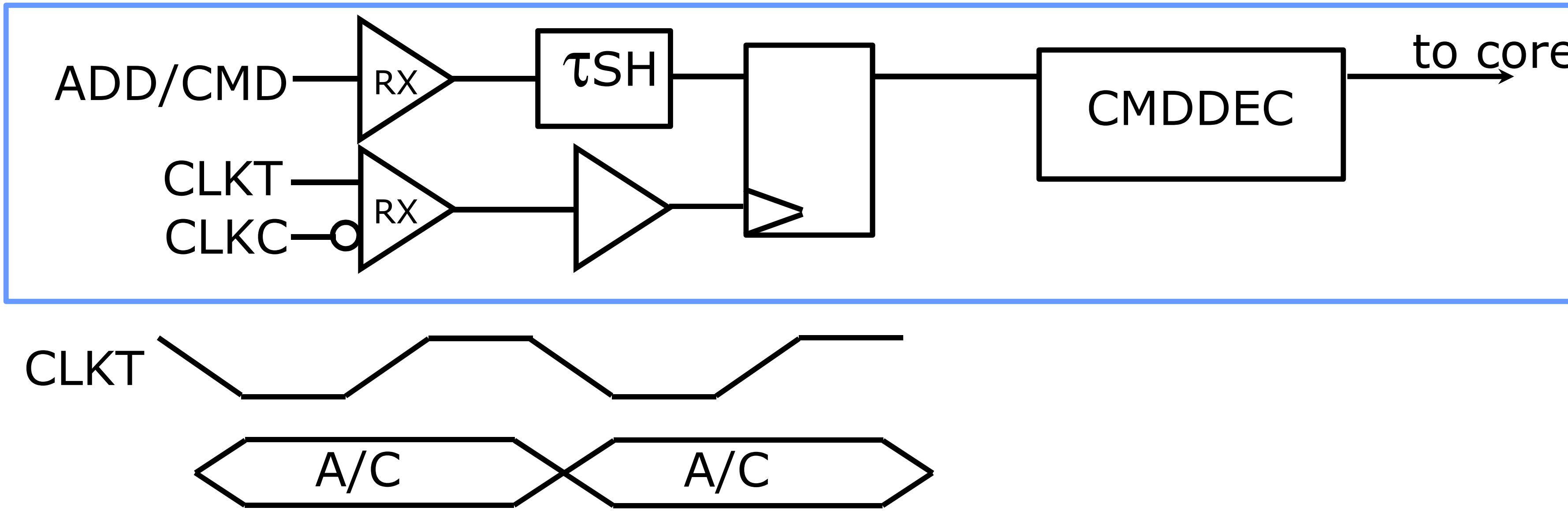
- Read FIFO guarantees read latency for controller

Hiding the Core: tCCD



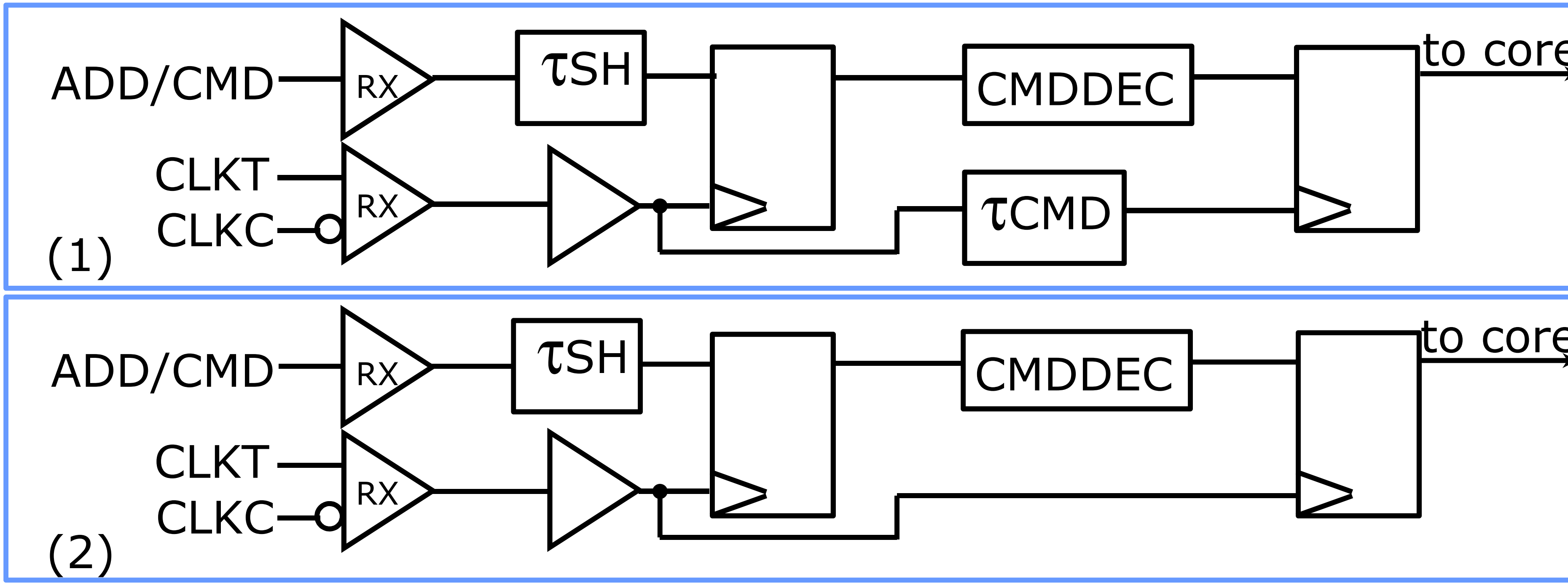
- N-bit prefetch to enhance per-pin data-rate to N / tCCD
  - Internal realization by multi-BL/COL and multi-COL activation
- N increased from 1 to 8 per DQ between SDR and DDR3

CMD/ADD System (1)



- Generic implementation
- Delay (τSH) to adjust setup/hold time at ADD/CMD-latch
  - (τSH) replicates delay of clock-buffering/distribution
  - For best replication, try to mimic design (gates, slew-rates)

CMD/ADD System (2)



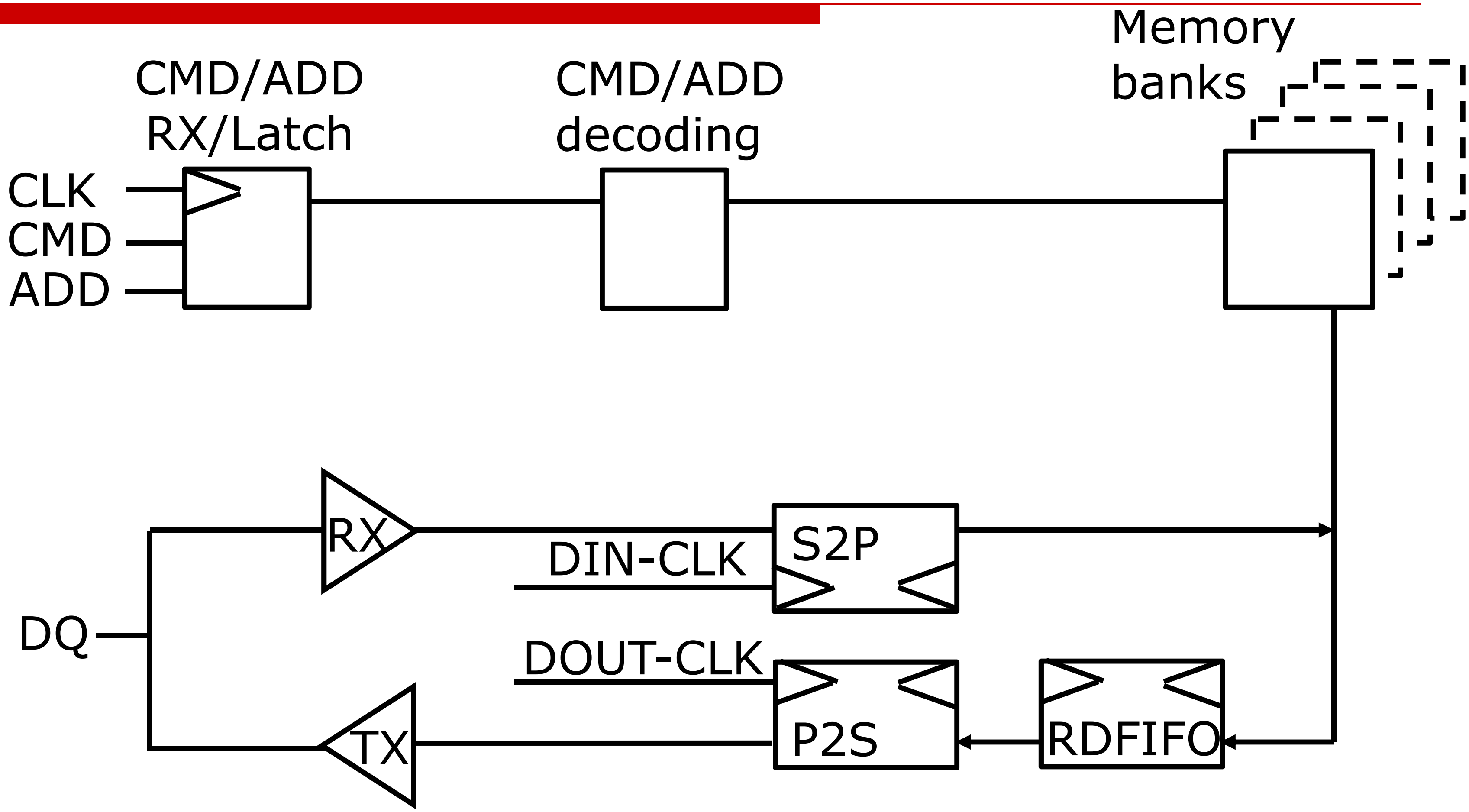
Wave-pipe style (1)

(τCMD) replicates CMDDEC-delay => no synchronous latency

Standard-logic style (2)

Follows logic design style, but 1 tCK synchronous latency

Functional Diagram



- DQ clocking is dependent on DRAM-family

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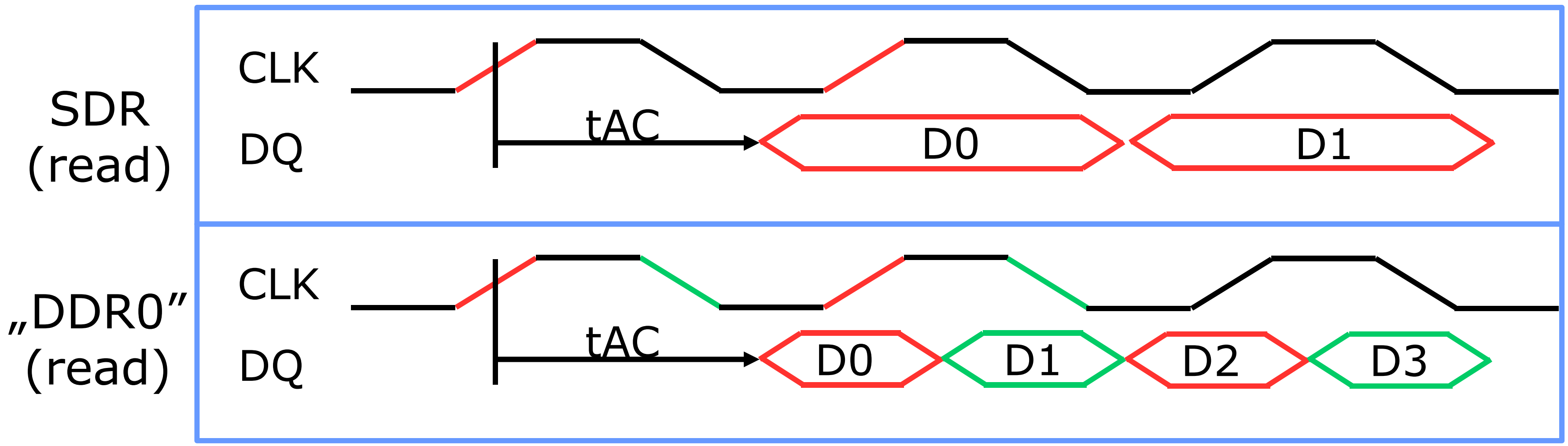
Basic DDR Concept

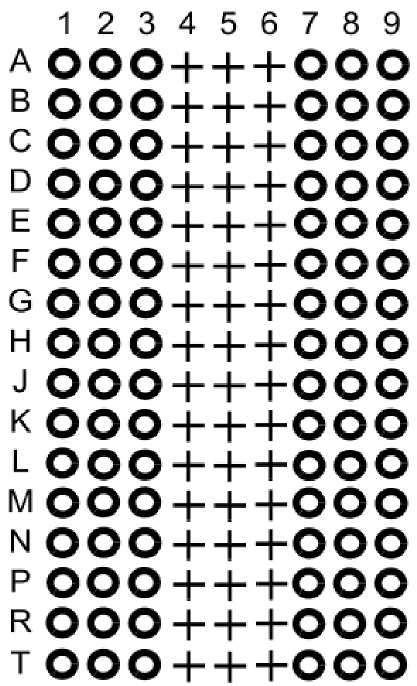


- Synchronous DRAMs evolved from SDR to DDR4
- Basic concept of DDR is to use both edges of the clock
  - Doubles data-rate for given CLK-frequency

DDR4

- Successor to DDR3 as main memory on modules
  - Evolution of DDR3-interface
  - Lower Vdd to 1.2V
  - Enable data-rates up to 3Gbps
- Design goals
  - Increase data-rate while keeping die-size in bounds
  - Server reliability
- Design features to support these goals include:
  - Data-Bus Inversion
  - Bank-group
  - IO-calibration
  - R A S

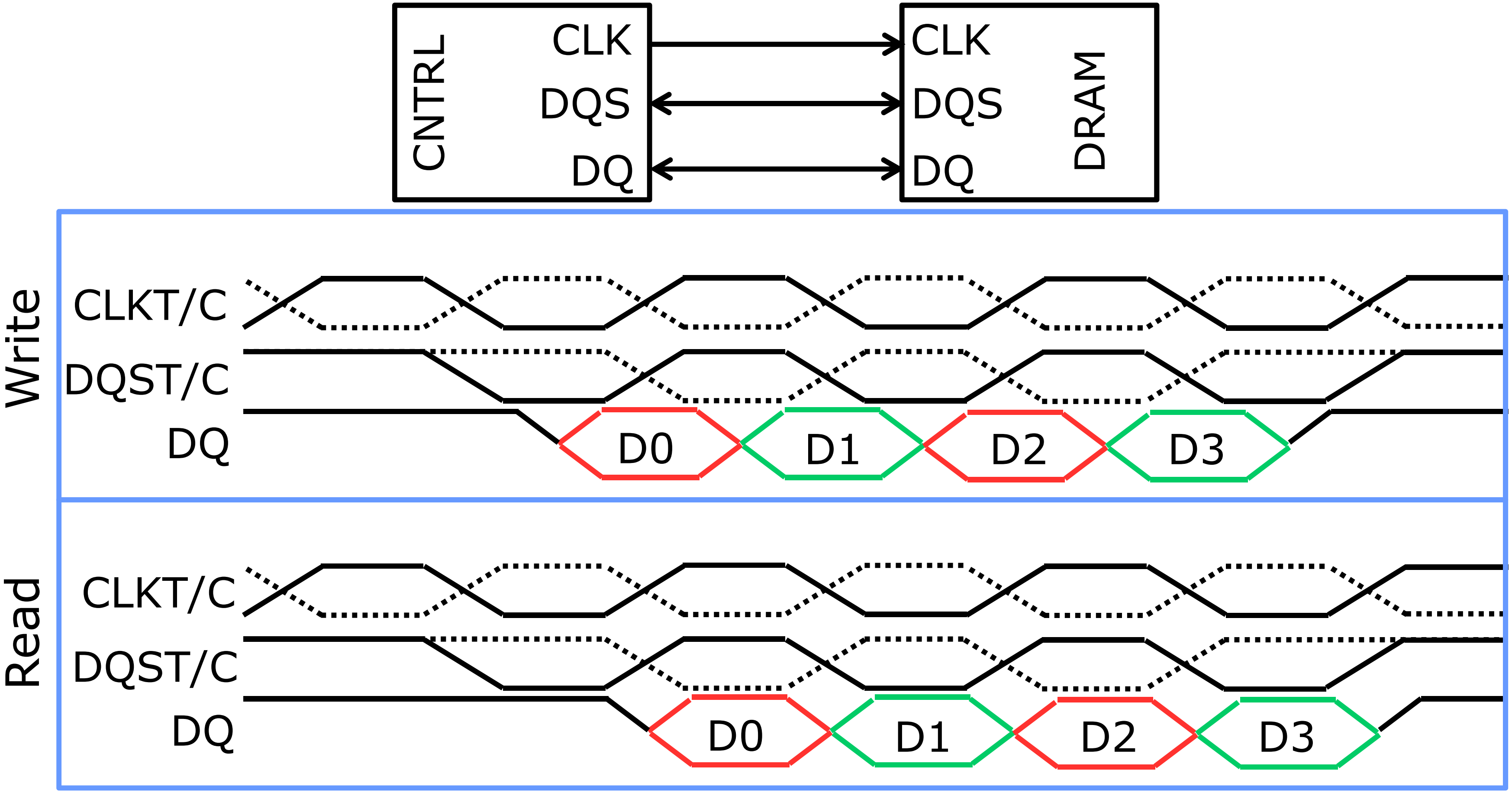
Ball Out



VDD	VSSQ	TDQS_c[3]		DM_n, DBI_n TDQS_t[2], (NC)[1]	VSSQ	VSS
VPP	VDDQ	DQS_c		DQ1	VDDQ	ZQ
VDDQ	DQ0	DQS_t		VDD	VSS	VDDQ
VSSQ	DQ4 (NC)[1]	DQ2		DQ3	DQ5 (NC)[1]	VSSQ
VSS	VDDQ	DQ6 (NC)[1]		DQ7 (NC)[1]	VDDQ	VSS
VDD	(C2)[5] ODT1[6]	ODT		CK_t	CK_c	VDD
VSS	(C0)[5] CKE1[6]	CKE		CS_n	(C1)[5] (CS1_n)[6]	TEN (NC)[7]
VDD	WE_n A14	ACT_n		CAS_n A15	RAS_n A16	VSS
VREFCA	BG0	A10 AP		A12 BC_n	BG1	VDD
VSS	BA0	A4		A3	BA1	VSS
RESET_n	A6	A0		A1	A5	ALERT_n
VDD	A8	A2		A9	A7	VPP
VSS	A11	PAR		A17 (NC)[4]	A13	VDD

- 0.8mm x 0.8mm physical ball pitch
- Outer data – Outer control
- 8 DQ example

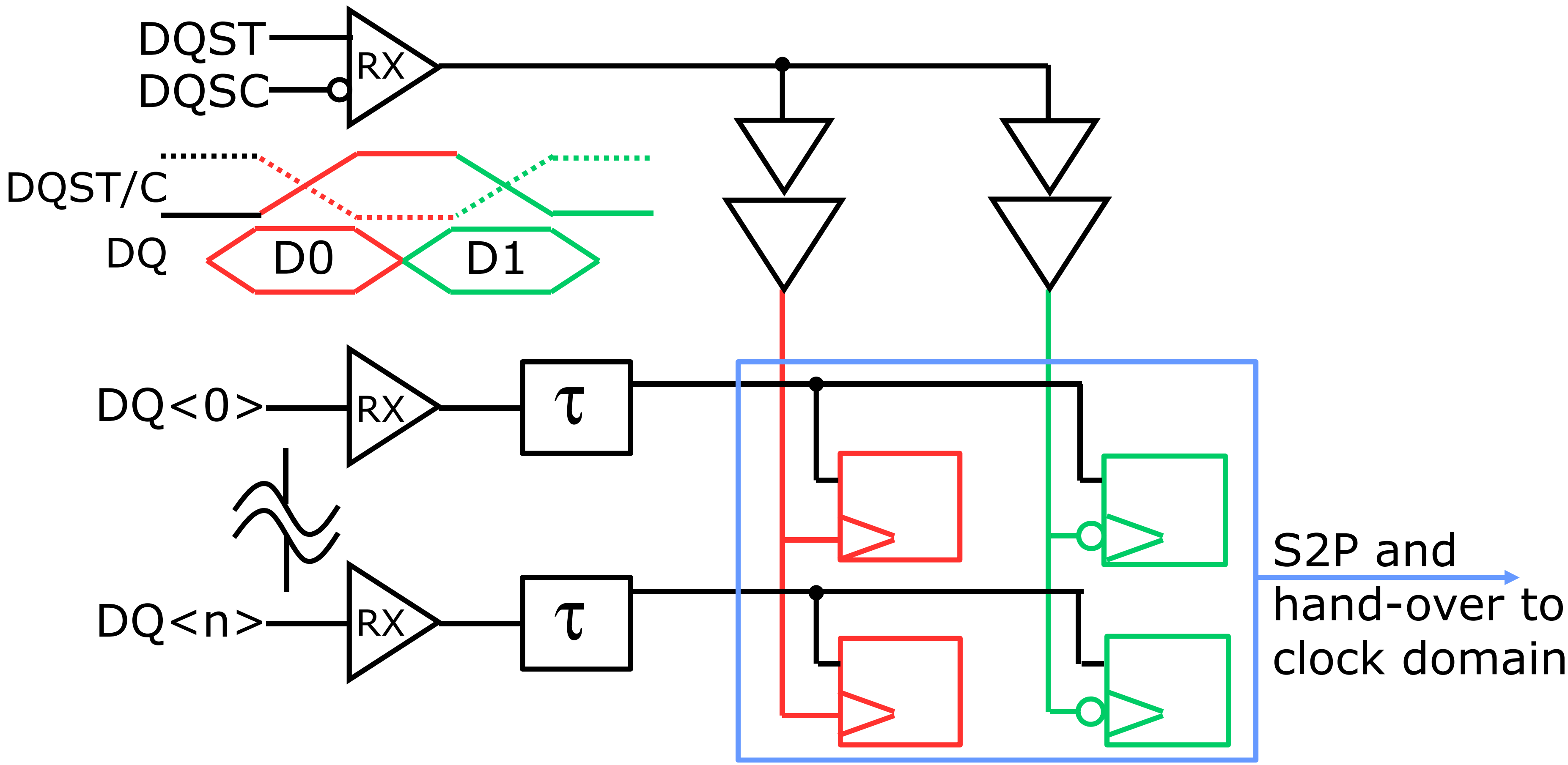
Data Strobe Concept



- Bi-directional strobe DQS synchronous to DQ
  - Center-aligned write and edge-aligned read

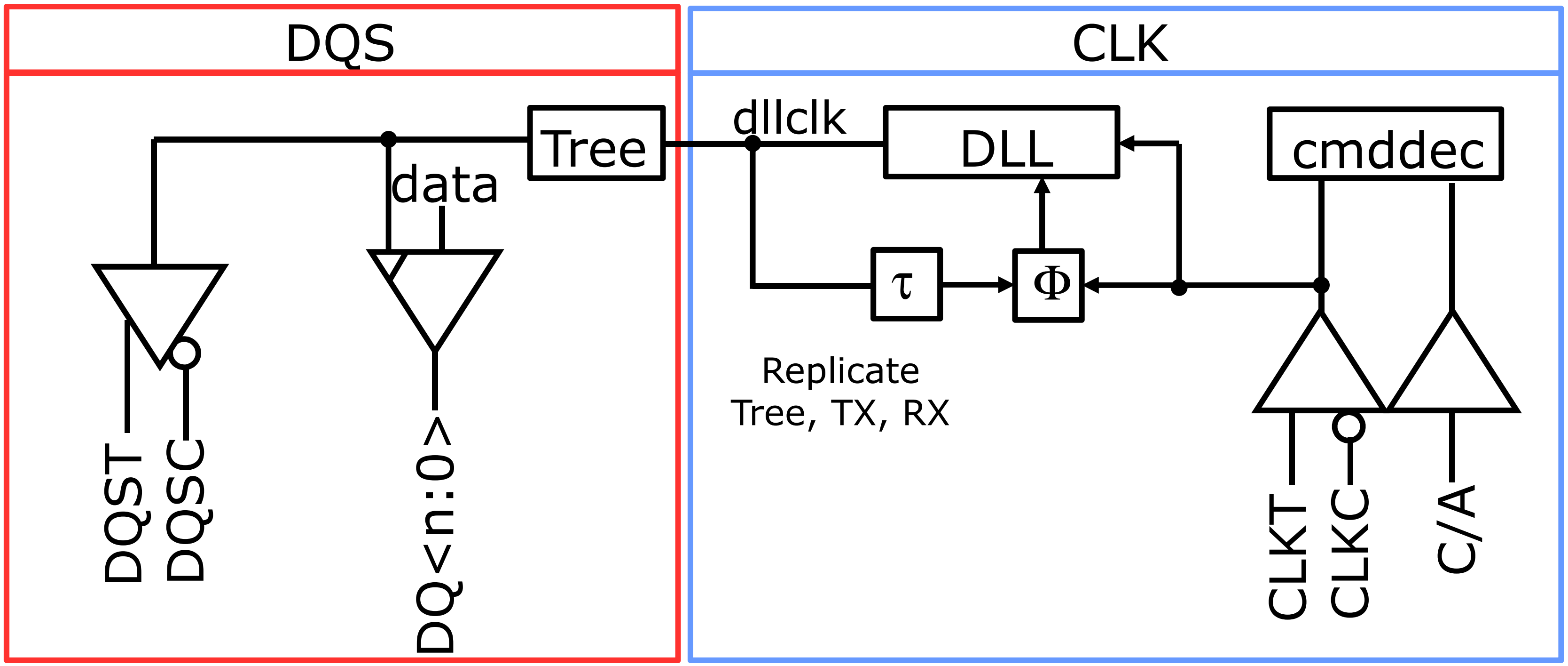
Example burst4

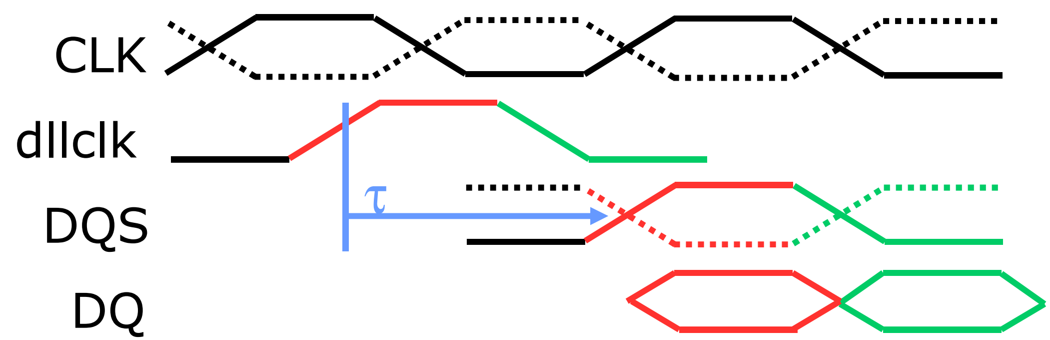
Clocking Scheme (Write)



- Controller drives DQS

Clocking Scheme (Read)



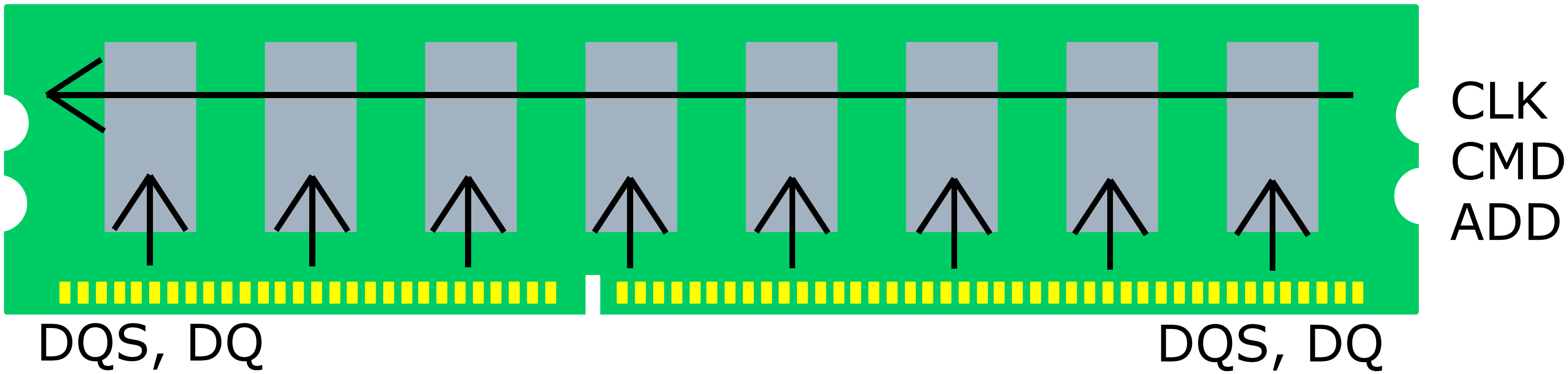


- DRAM drives DQS

Clocking Scheme

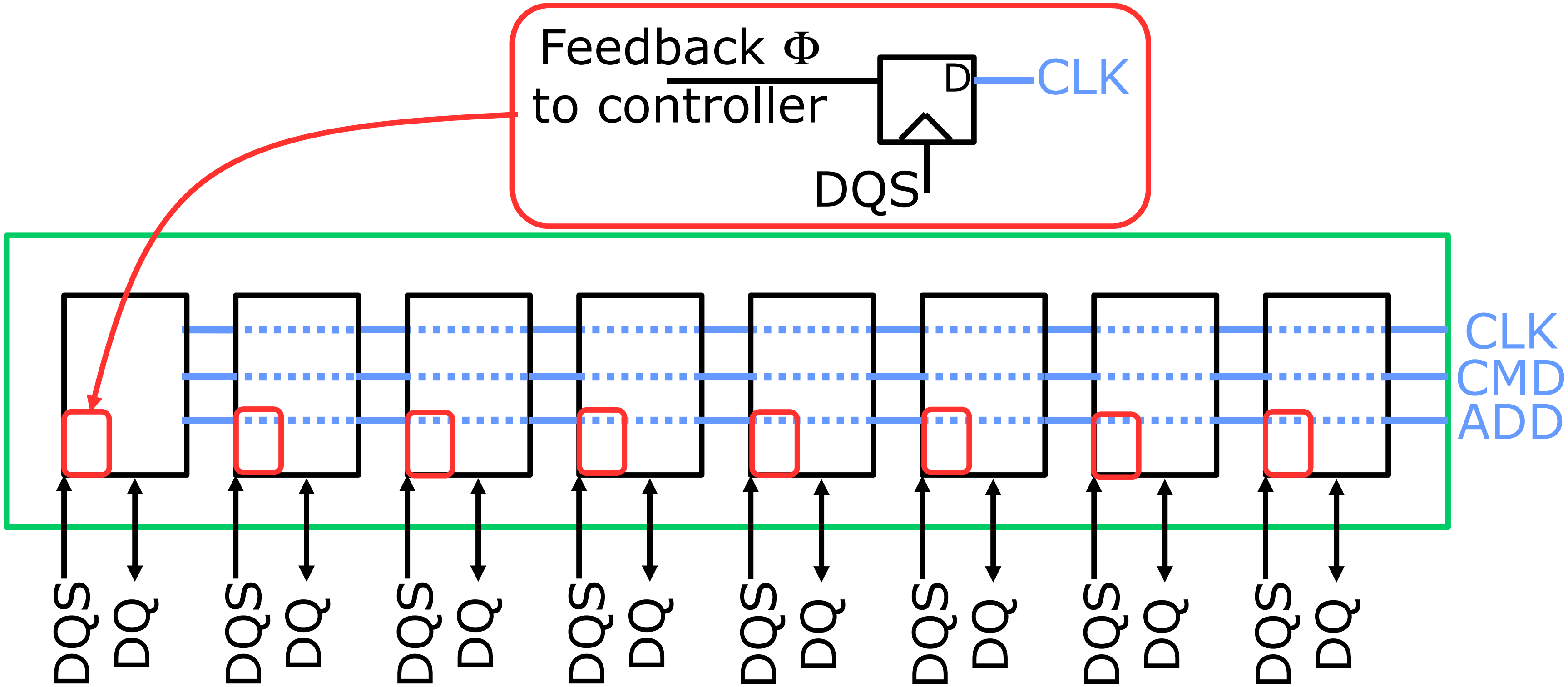
- Read-Alignment by DLL to global system clock
  - Different memories interchangeable
  - DLL and clock distribution adds jitter
- Jitter on DQ and DQS correlated
  - Advantage for DQS-clocked receiver
  - Delay replica in DQ-path may add jitter
- Correct write relies on timing between DQS and CLK
  - Command interface operates in CLK domain
  - DQ interface operates in DQS domain

Module-Level Clocking (1)



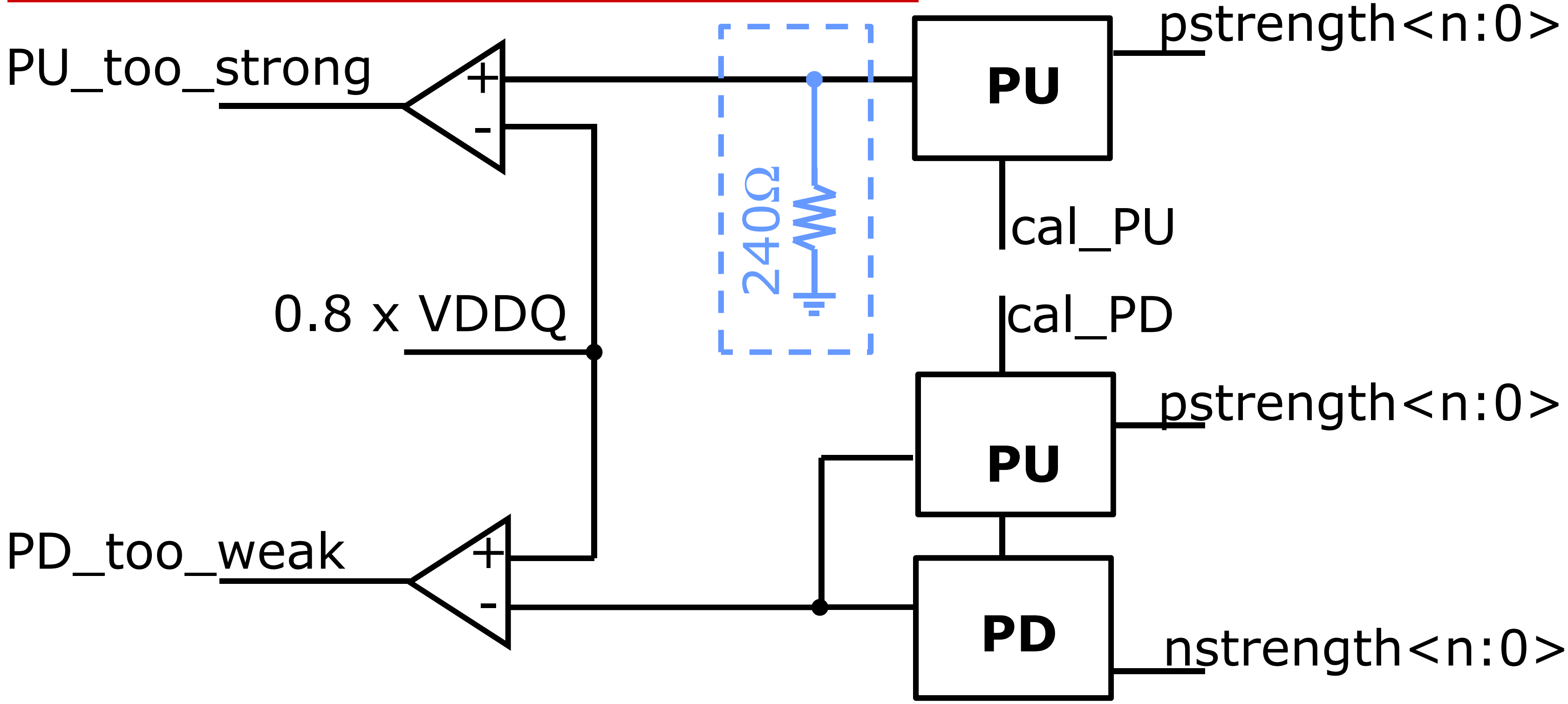
- DDR4 modules employ fly-by CMD/ADD architecture
  - Efficient architecture to distribute command/address
  - Skew between CLK and DQS depends on physical position
- DRAM requires timing between CLK and DQS (tDQSS)
  - Enables correct handover of data from DQS to CLK domain

Module-Level Clocking (2)



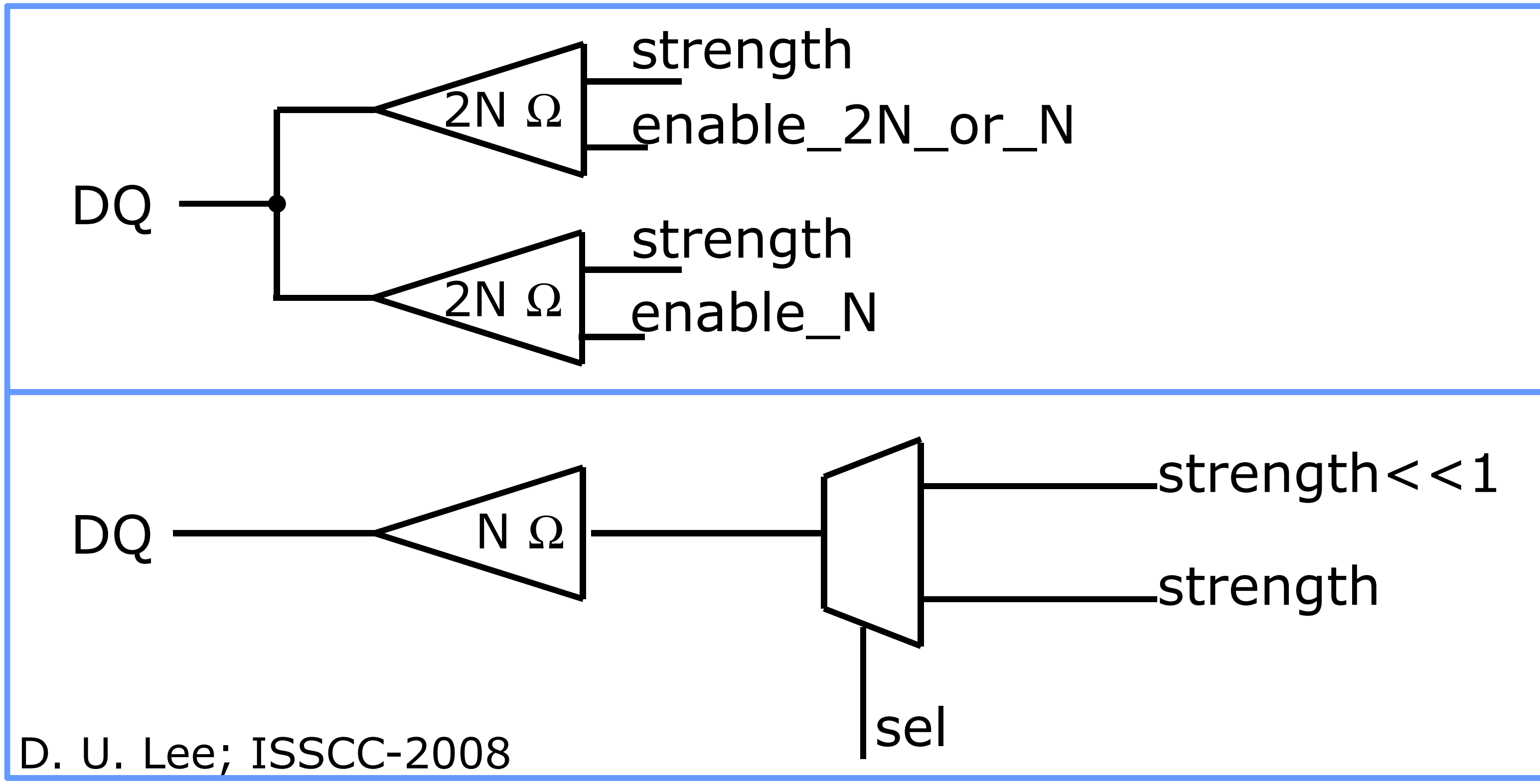
- Adjust DQS/CLK phase difference from controller side
- Memory adds phase detector to find optimum setting

IO Calibration (1)



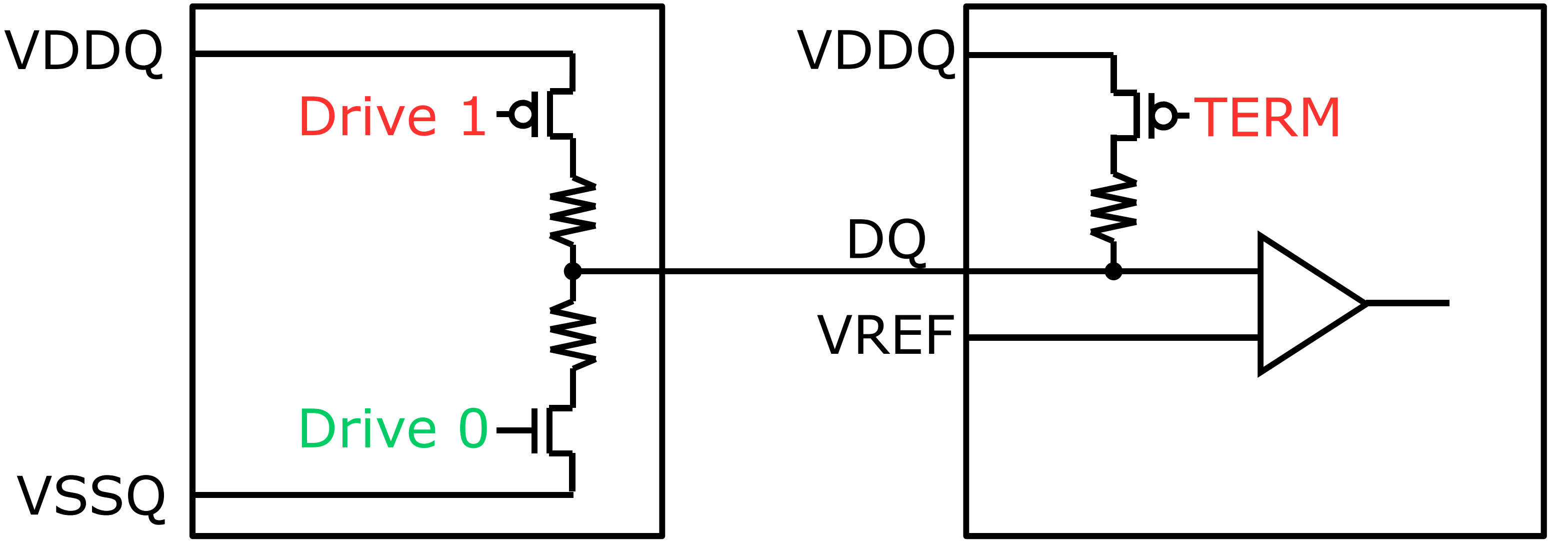
- Provide external reference resistor for calibration
  - calibrate **P**ull**U**p against external resistor (“ZQ-resistor”)
  - calibrate **P**ull**D**own against [calibrated] **PU**
- Driver-strength is typically binary-coded

IO Calibration (2)



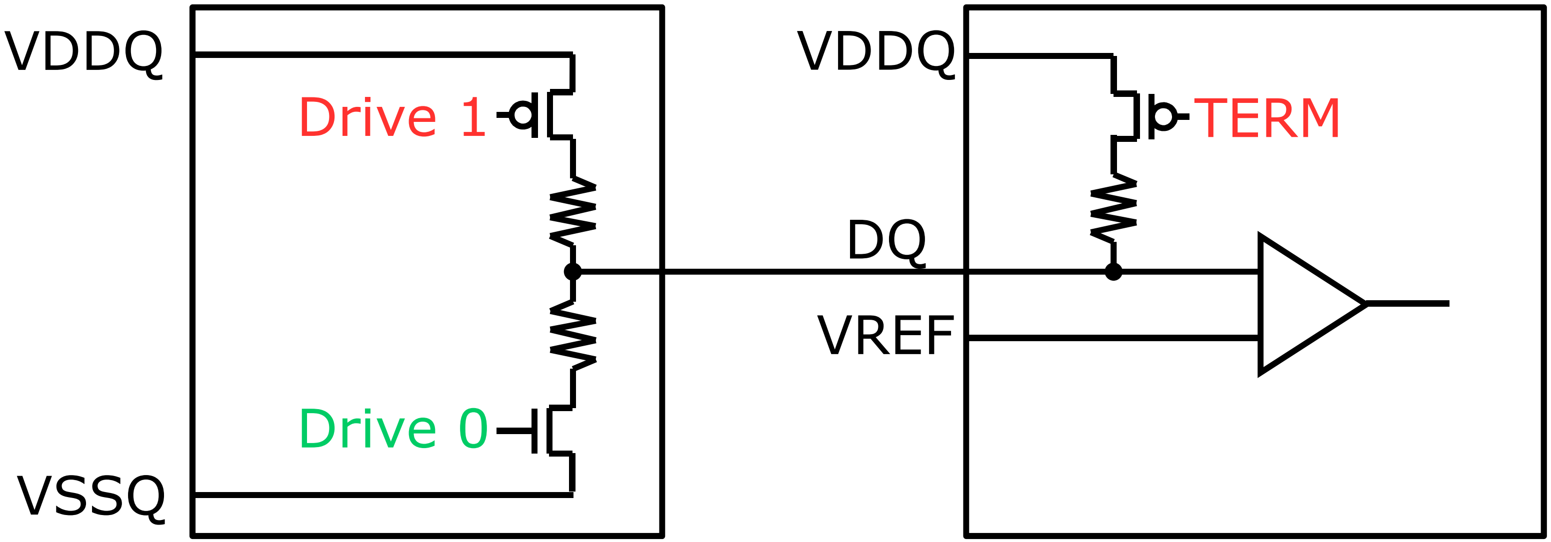
- Most specifications require multiple drive strengths
- Implement by sectioning or binary-shifting

Signaling (1)



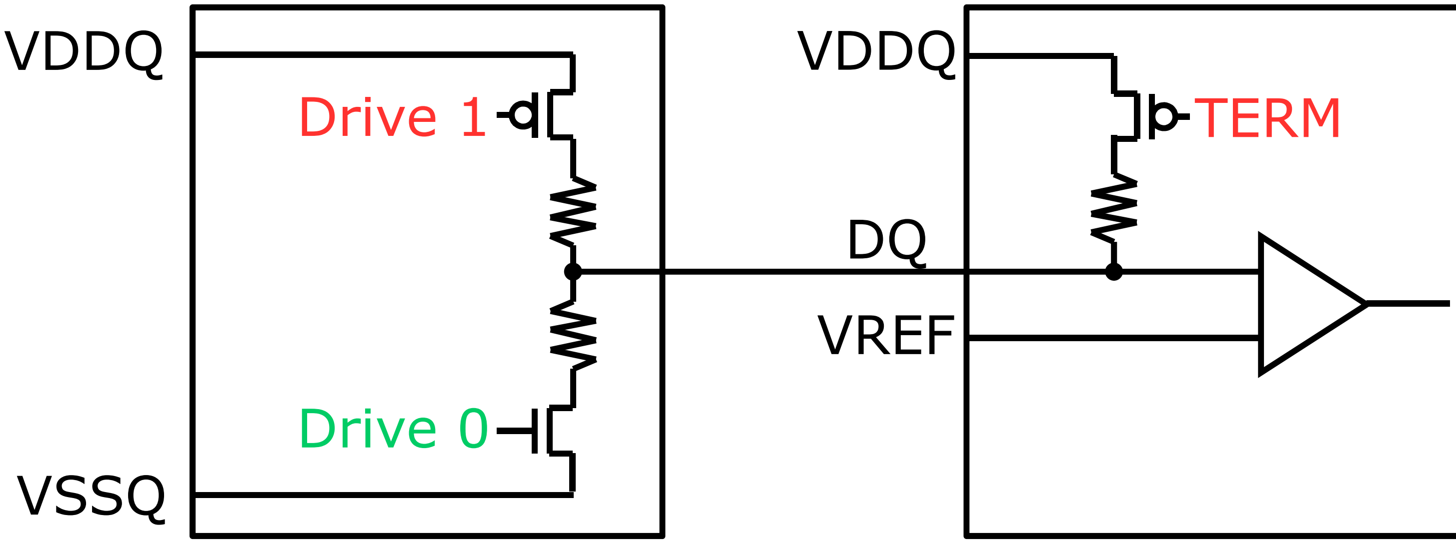
- DDR4 follows a single-ended signaling scheme
  - Smaller pin-count compared to differential signaling
  - Only DQS and CLK are differential signals
- Providing VREF enables pseudo-differential RX
  - Reference voltage is nominally at midpoint of data-eye

Signaling (2)



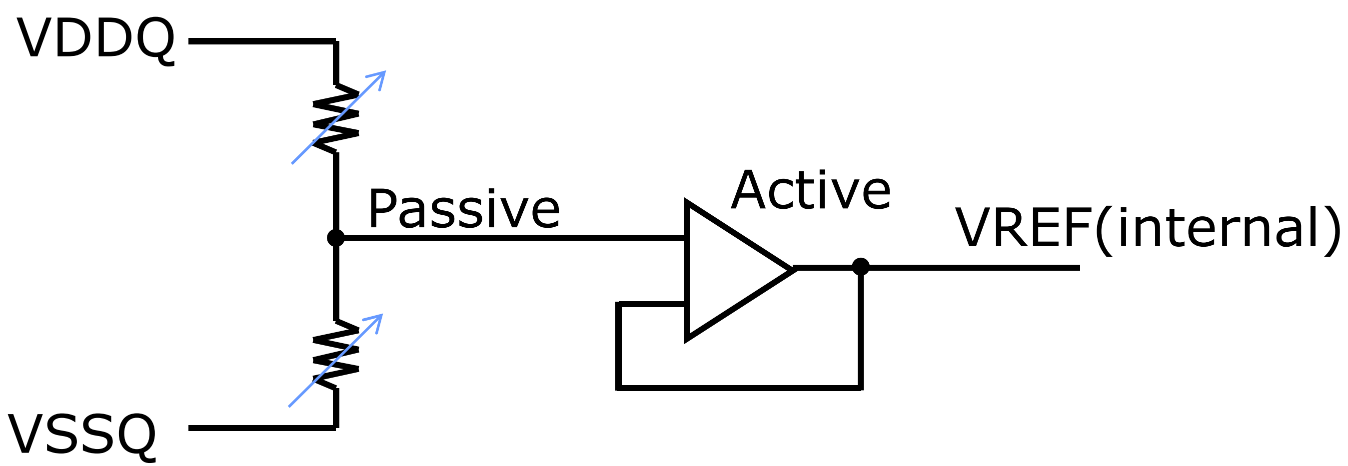
- On-die termination (ODT) to minimize reflections
- DDR4 employs high-tap termination
  - Termination and '1'-driver are same pMOS/resistor
  - Minimizes input capacitance
- Series resistors improve linearity of driver and termination

Signaling (3)



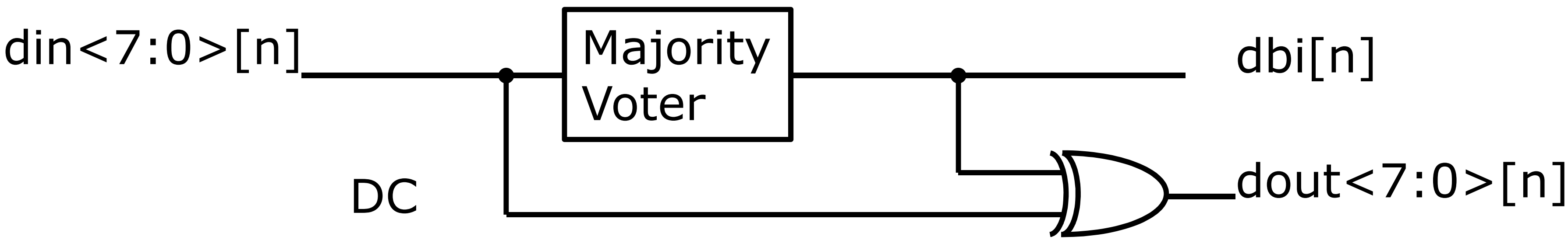
- Driver and termination impedance are programmable
  - Allows to adapt to different system environments
  - Values between 240Ω/1 and 240Ω/7 possible (sectioned)
- V(mid-point) > 0.5 x VDDQ enables nMOS-only receiver

VREF Generation



- Classical scheme: provide VREF externally by system
- Alternative scheme: generate VREF on the die
  - Both active and passive schemes possible
- On-die generation reduces component count in the system
- On-die generation enables easier trimming
  - Programmable on-die VREF-generator
  - Adjust VREF to ideal VREF as determined by data-eye

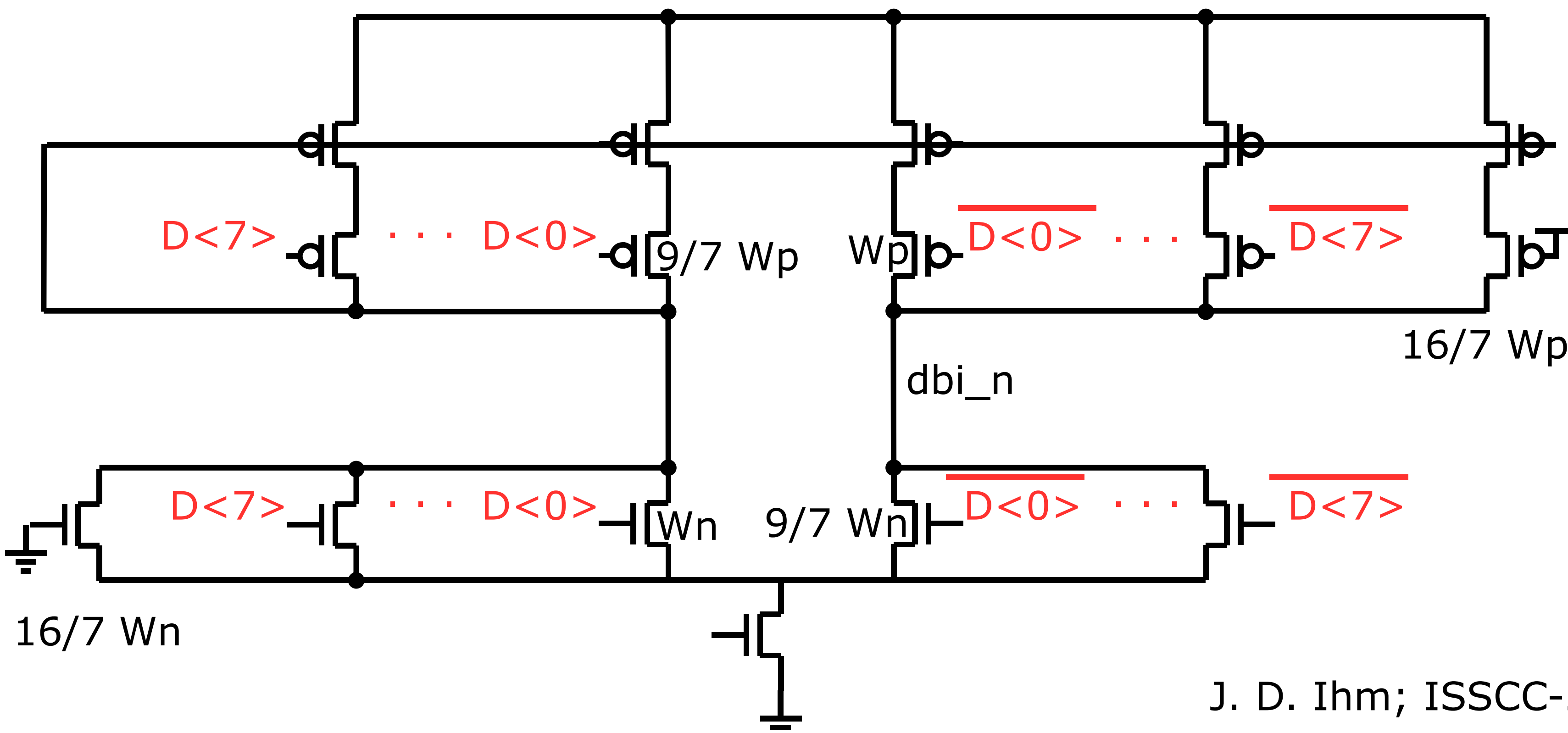
Data Bus Inversion (DBI)



K. Nakamura; VLSI-1996
J. D. Ihm; ISSCC-2007

- DDR4 is high-level terminated
  - Driving statically a ‘0’ requires a constant current flow
- Add one DBI-bit to one byte of data
  - DBI: Data-bus-inversion
- Dependent on DBI-bit, byte will be inverted at receiver
- Two flavors of DBI in use
  - DC: Limit the number of states
  - AC: Limit the number of transitions

DBI: Analog Majority Voter



J. D. Ihm; ISSCC-2007

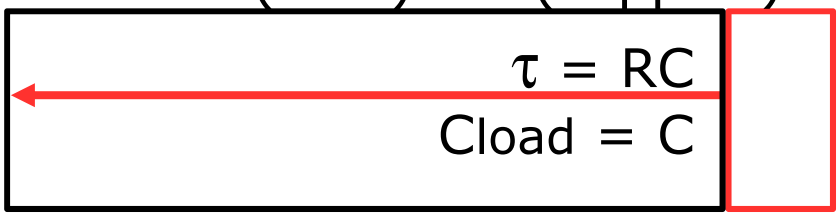
- Optimized analog voter resolves 4/4 indecision by sizing
- Fast operation achieved at the price of analog sensitivity

DBI: Digital Majority Voter

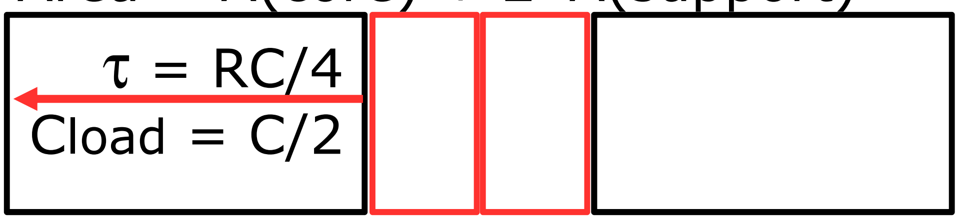
- Many circuit possibilities are available
  - Another possibility is logic-synthesis from RTL
- Decide for which parameter to optimize
  - Power dissipation
  - Maximum propagation delay
  - Variation of propagation delay

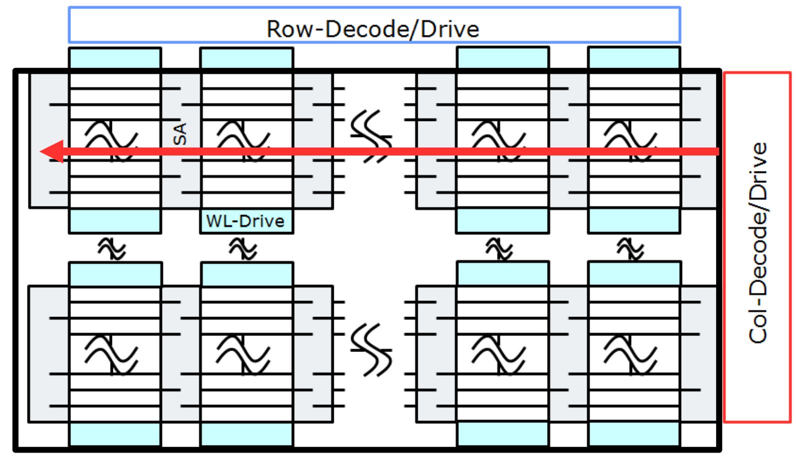
Bank Group Concept (1)

Area = A(core) + A(support)



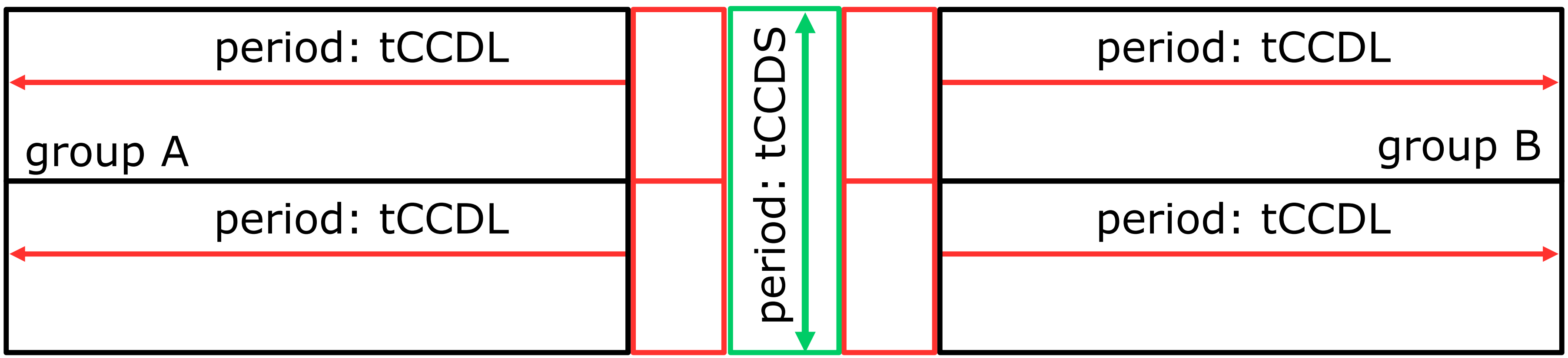
Area ~ A(core) + 2*A(support)

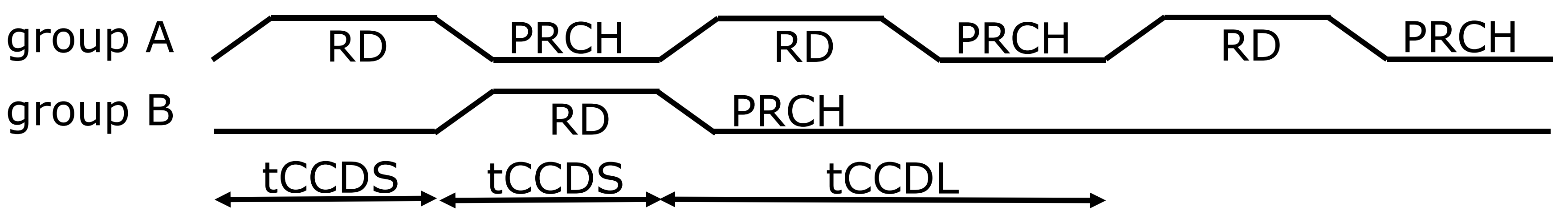




- Column speed of the core is limited by RC-delays
  - A fast core is conceptually easy => limit length of line
  - But expensive in area => added decoders, drivers, …
- Compromise to combine slow core with fast data-path

Bank Group Concept (2)





- Partition memory into groups
  - tCCDS: Valid for column accesses outside group
  - tCCDL: Valid for column accesses inside group

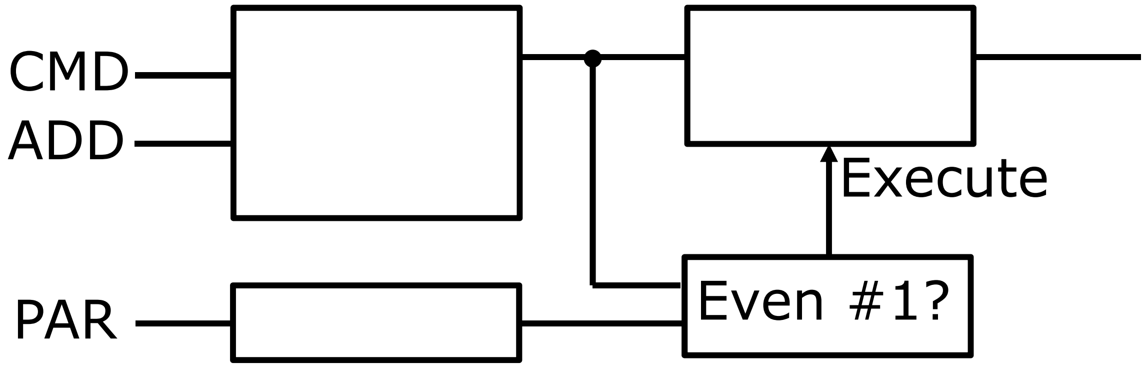
R A S features

- DDR4 is targeted at servers
- DDR4 includes features that improve **R**eliability, **A**vailability, **S**erviceability (RAS)
  - Post-package repair on devices in the field
  - Parity protection on command/address bus
  - CRC protection on data-bus

Post-Package Repair

- Every DRAM carries redundancy
  - Used in wafer/component processing to reach 0% fails
- Redundancy selection
  - Historically by laser fuse-cut
  - Rapidly shifting to electrical fuse-cut
- Standardized protocol provided for in-system repair
  - Very limited number of repairs allowed

CA Parity protection



- Parity bit is sent by controller for CMD/ADD bus
- In case of parity error
  - Command execution is blocked
  - Parity error is reported back to the controller

CRC protection on data-bus

- Checksum protects data-transmission on DQ-bus
- In general, CRC is possible during writes and reads
- DDR4 provides CRC only during writes
  - Controller adds CRC-bit at the end of the burst
  - Protects data written to memory
  - Read data protection can be handled on system-level

CRC: Data Mask Problem

Transmission Pass

CMD	Mask	Data	Cell
WRT	No	D	D
WRT	Yes	$\overline{D}$	D

Transmission Fail

CMD	Mask	Data	Cell
WRT	No	D	D
WRT	Yes No	$\overline{D}$	$\overline{D}$

- Incorrect write can be recovered by repeating write
- Incorrect masked-write is non-recoverable
  - Contents of the memory cell is destroyed
- DRAM compares data to CRC. In case of error:
  - Write: no need to block write. Flag error to controller
  - Write w\mask: need to block write. Flag error to controller

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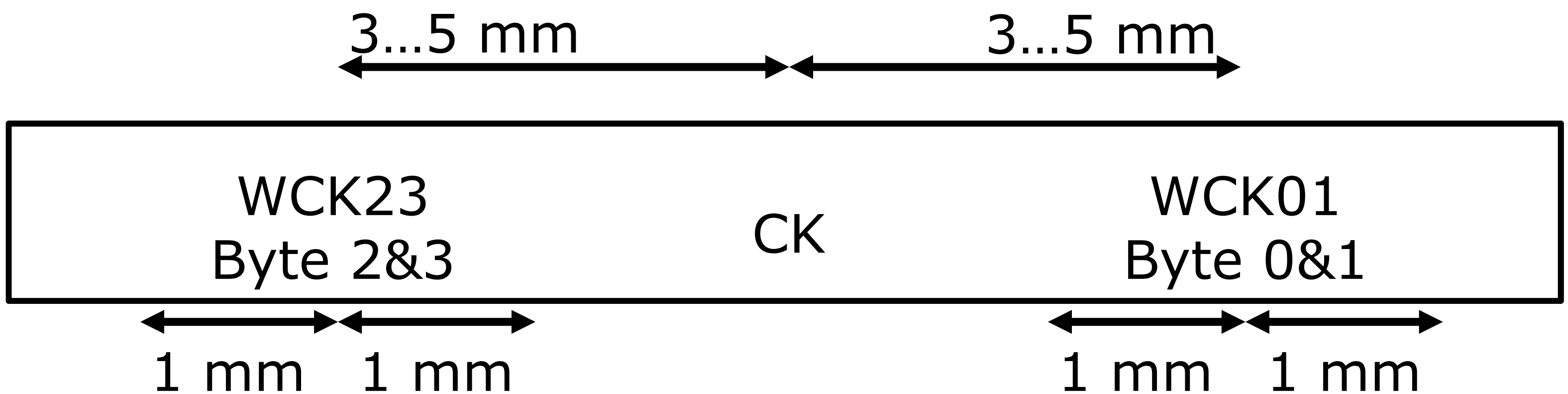
GDDR5

- Design goals
  - Bandwidth target originally 5Gbps, meanwhile at 8Gbps
  - Soldered down to eliminate connectors in bus
- GDDR5 introduced some techniques of DDR4
  - High-level termination
    - Implemented as 40Ω driver / 60Ω termination
  - Programmable VREFD
  - DBI
- Additional design features to enhance data-rate
  - Clocking scheme
  - Training
  - Improved RX/TX architecture
  - Optimized EDC concept

Ball-out

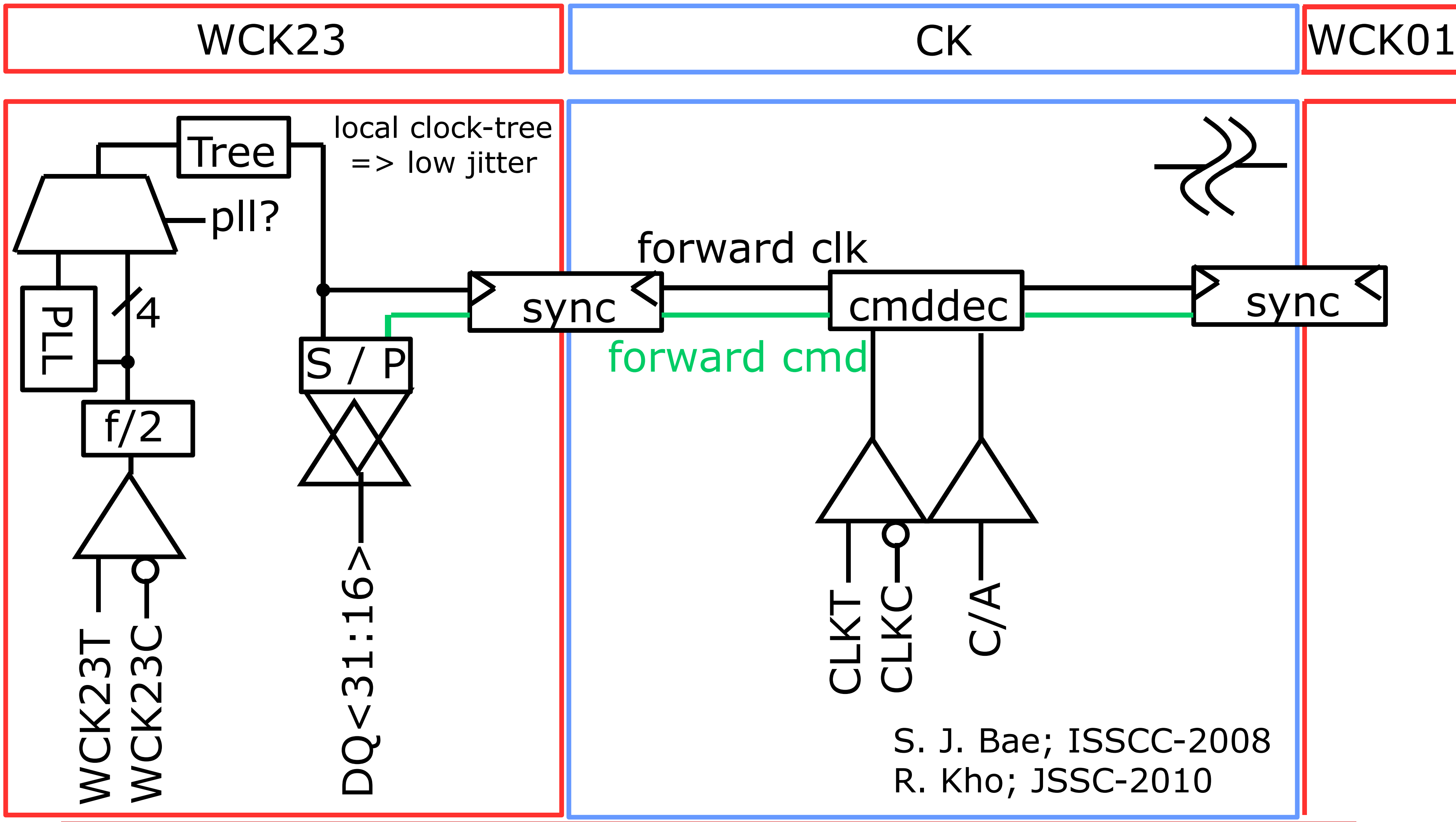
- 0.8mm x 0.8mm physical ball pitch
- Outer data – Inner control

Clocking Scheme (1)

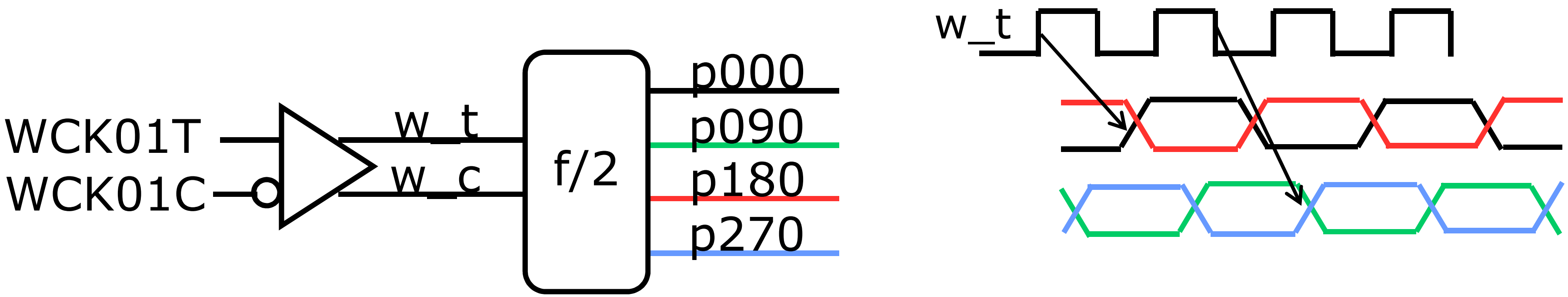


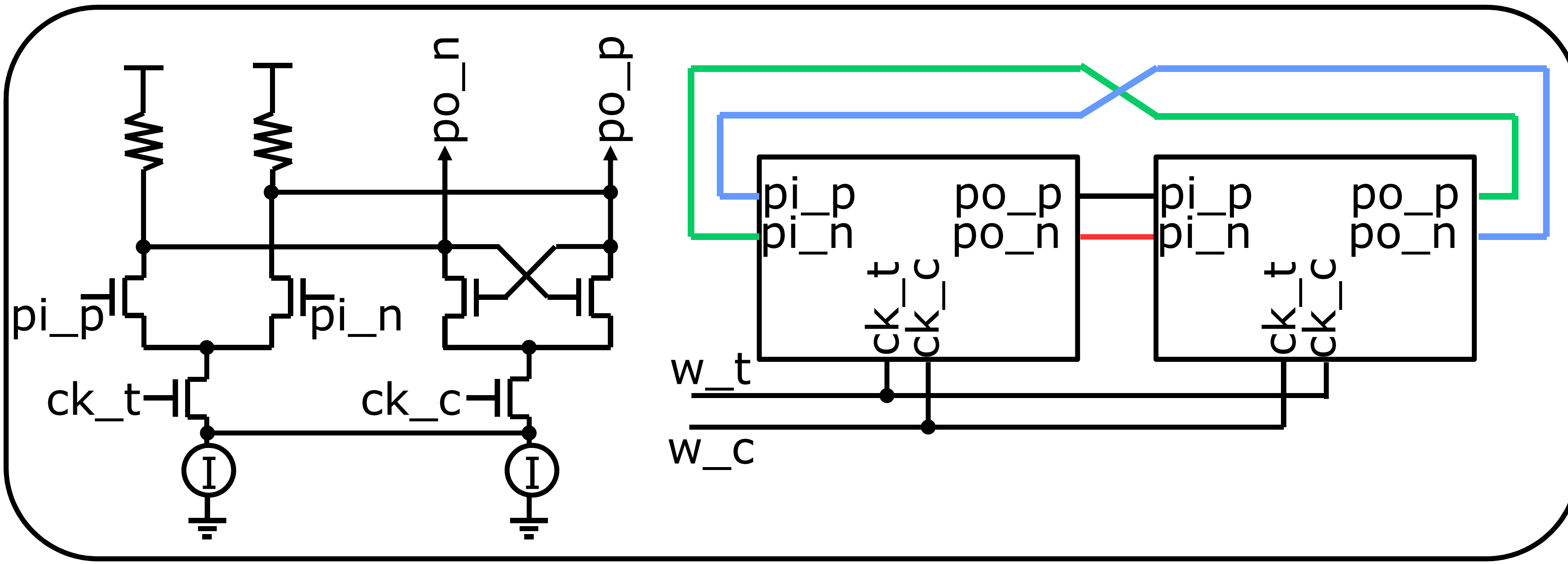
- GDDR5 introduces a pair of dedicated clocks for RX/TX
- Clock is local to two bytes
  - External WCK runs at twice the command clock frequency
  - Low-jitter, compact clock distribution
  - Clock may be distributed using CML
- WCK/CK phase alignment during initialization
  - DRAM adds phase detector to feedback phase to controller

Clocking Scheme (2)

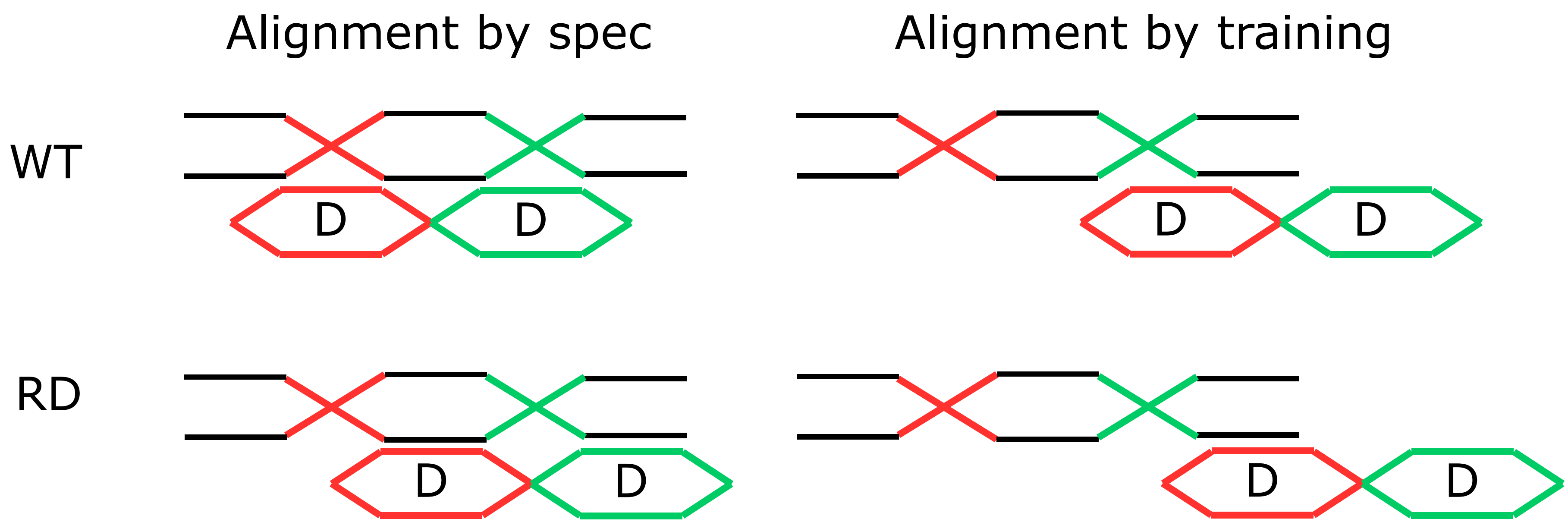


Clocking Scheme (3)





Training (1)

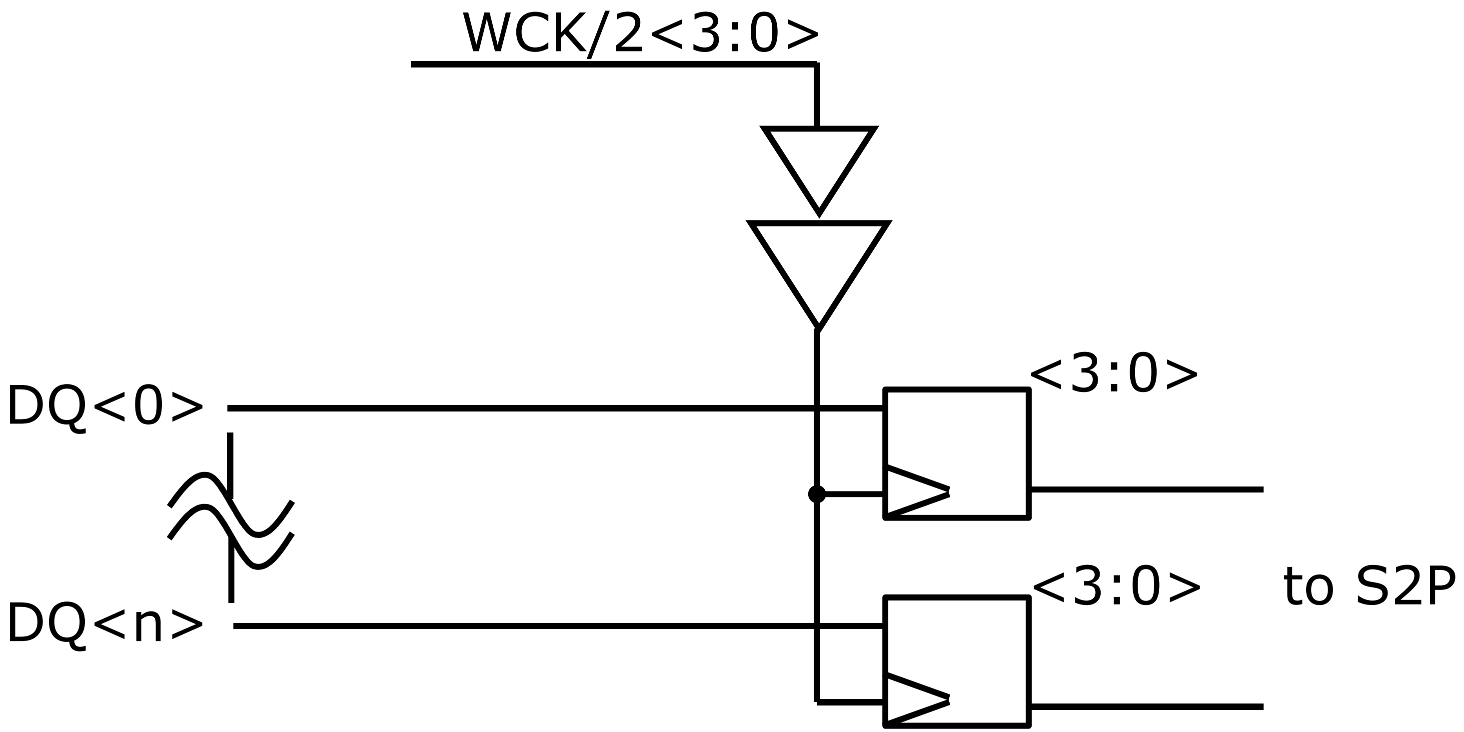


- Specification: Specification guarantees timing
  Defined RX/TX timing => DDRn
- Training: System drives data at optimum time
  Controller needs to search the data

Training (2)

- Requires new commands to initialize the chip
- Typical sequence
  - Introduce low-speed write to preload reliable data
  - Train reading by searching for preloaded data
  - Train writing using stable read
- GDDR5 implements initial preload via address-bus

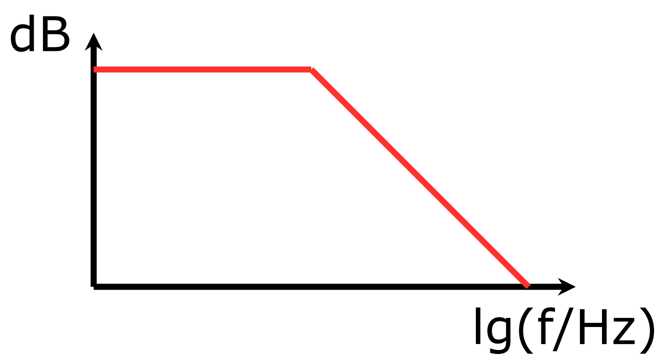
RX Architecture (1)



- Four samplers directly attached to pad
  - Potential penalty of increased pin capacitance

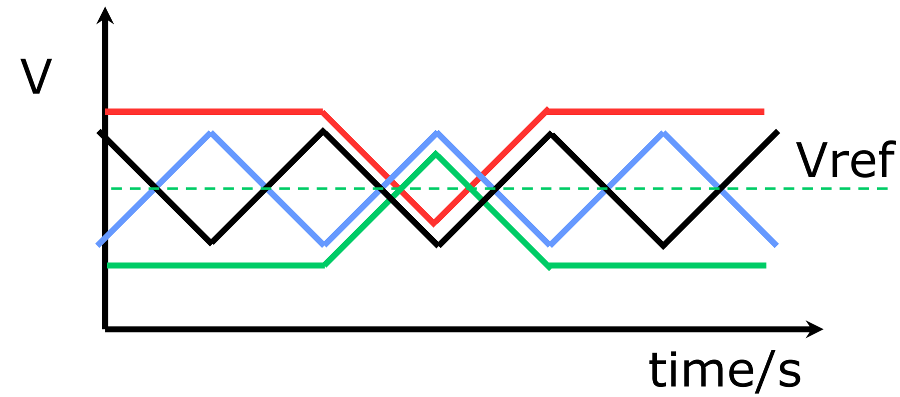
RX Architecture (2)

Channel attenuation



Channel length, material
Pin capacitance
Driver bandwidth

Inter-symbol interference

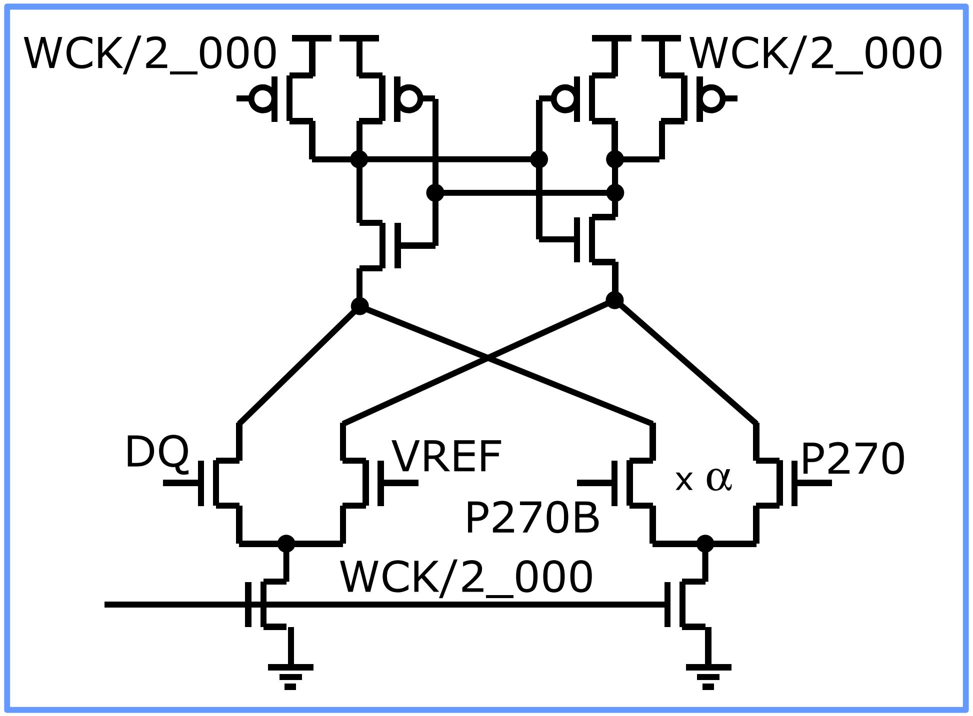


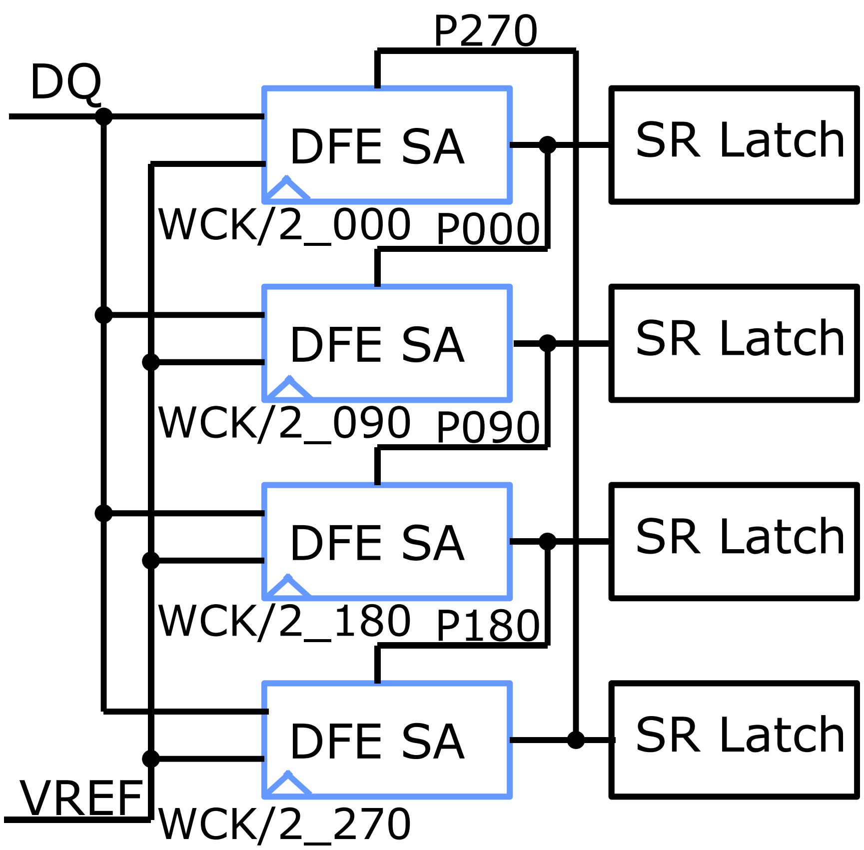
Slew-limitation

- Data rate of GDDR5 is limited by quality of signal at RX

RX Architecture (3)

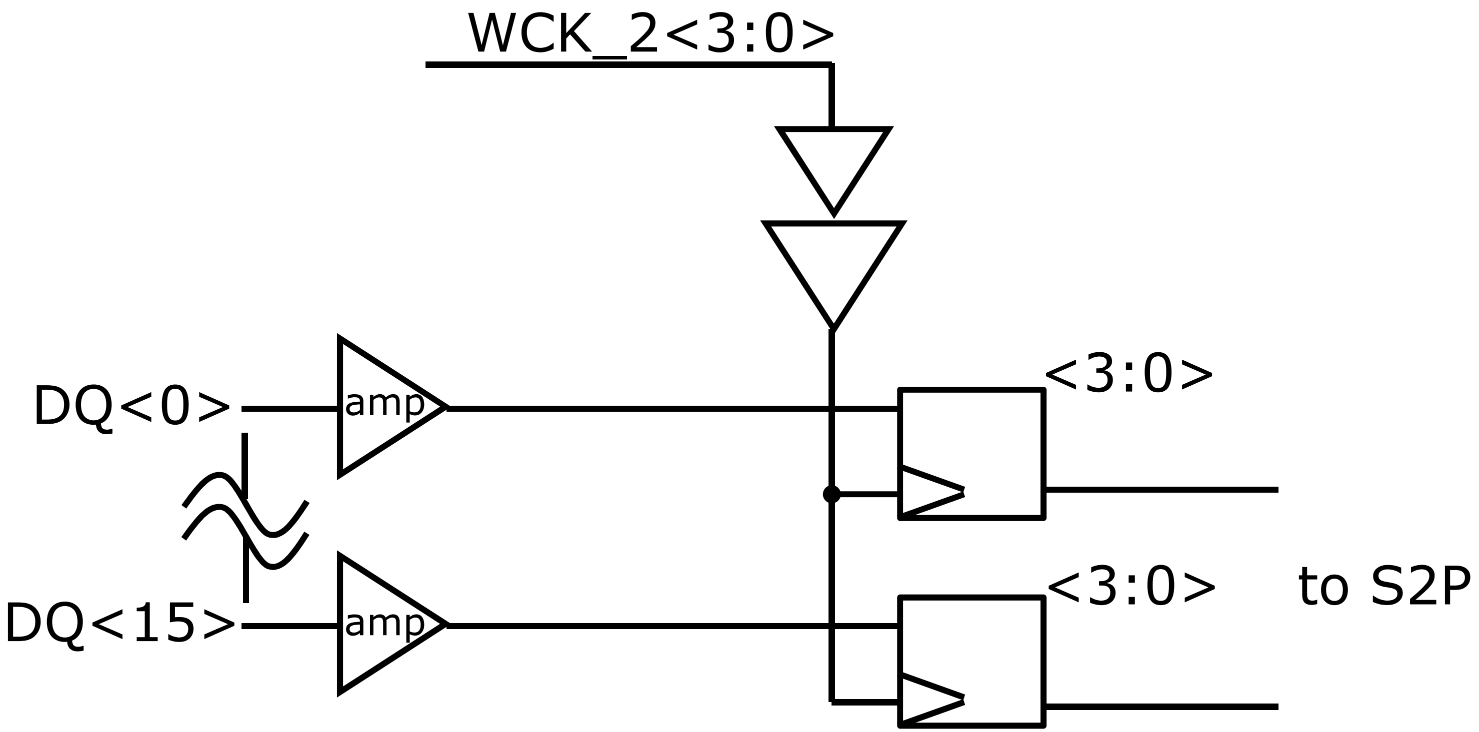
S. J. Bae; ISSCC-2008





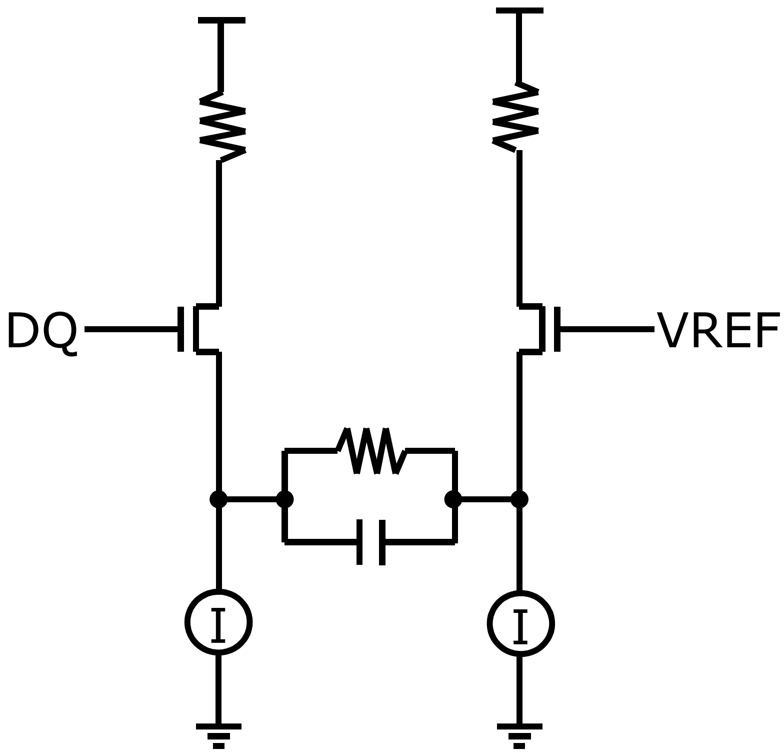
- Digital equalization to improve sampling window

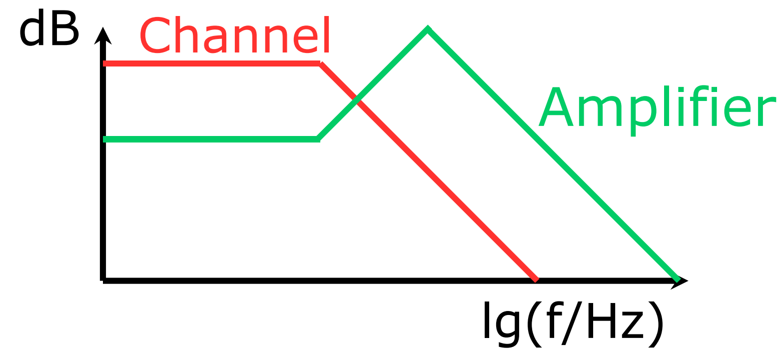
RX Architecture (4)



- Minimizes pin-capacitance by decoupling samplers
- Enables use of analog equalizer

RX Architecture (5)

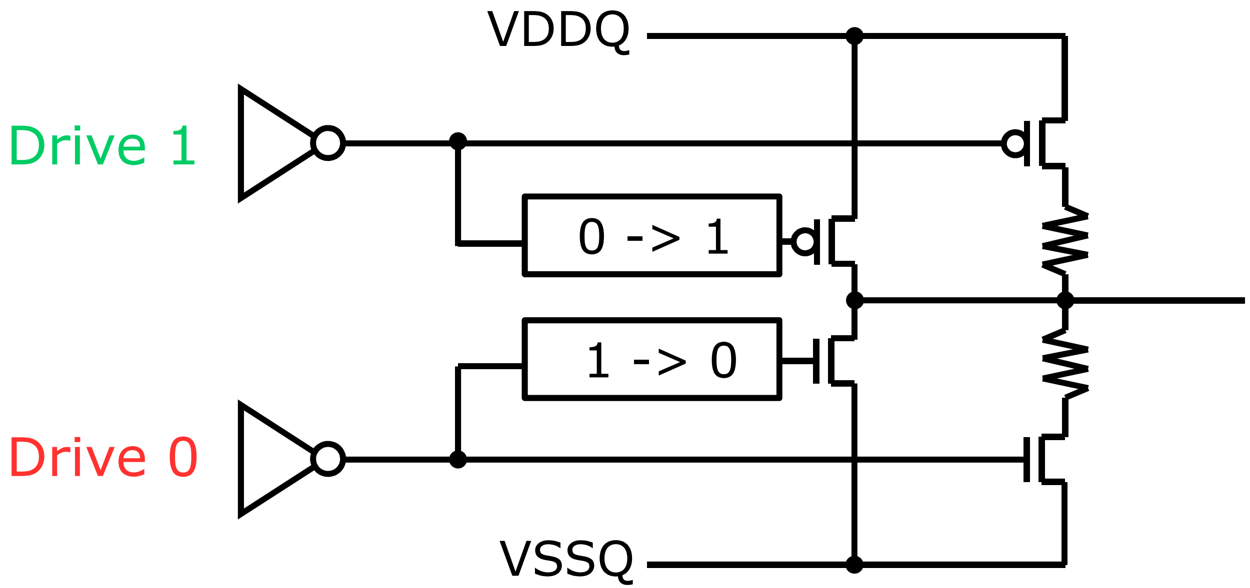




Y. Tomita; JSSC-2005

- Possible implementation for an analog equalization

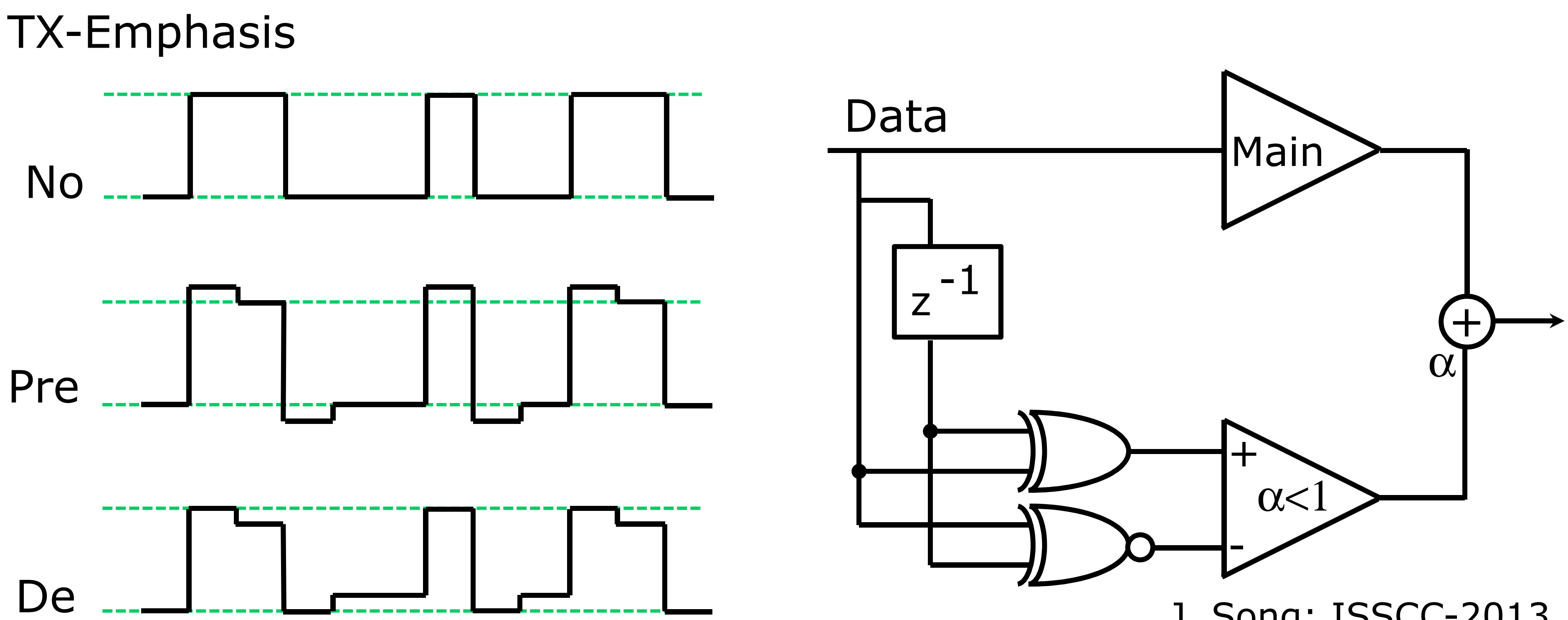
TX architecture (1)



Kho; ISSCC-2009

- Circuit transiently boosts PU/PD at first transition
  - Added transistor driven by edge-detecting circuit

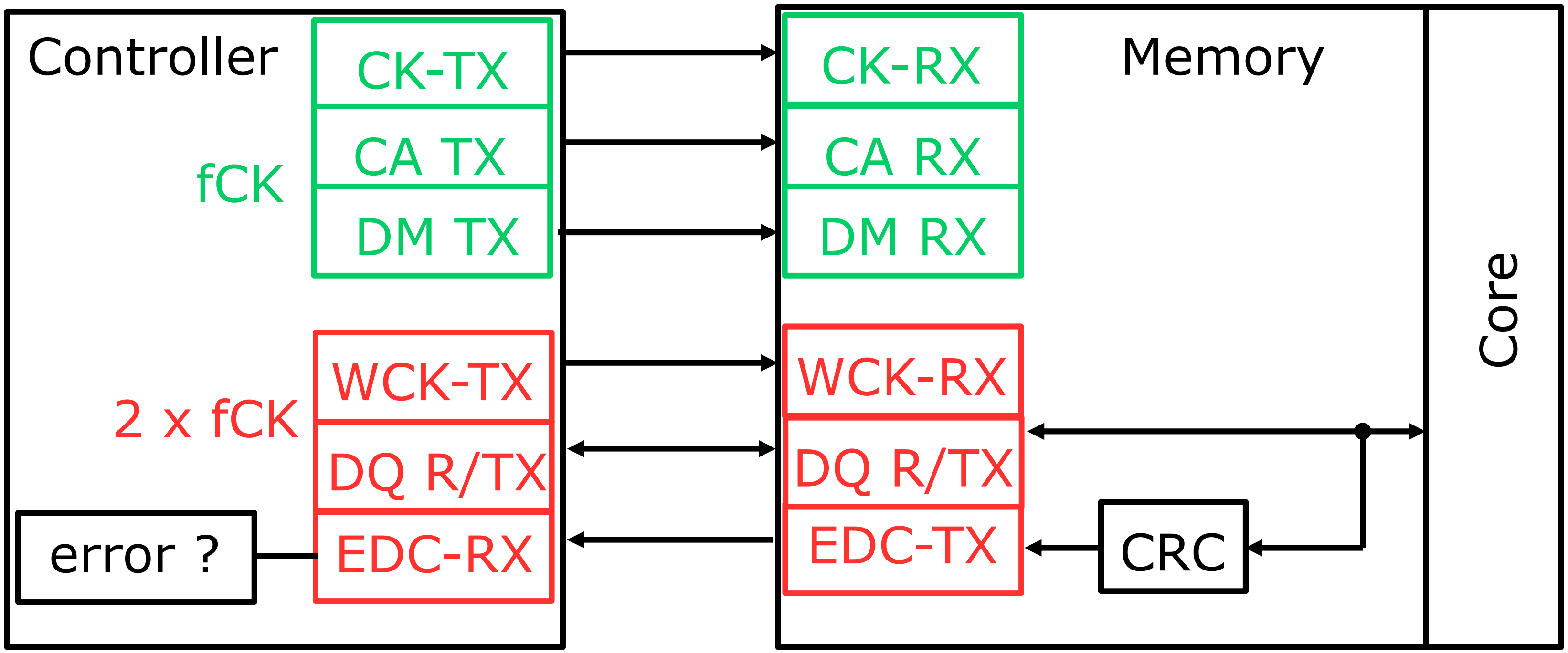
TX architecture (2)



J. Song; ISSCC-2013

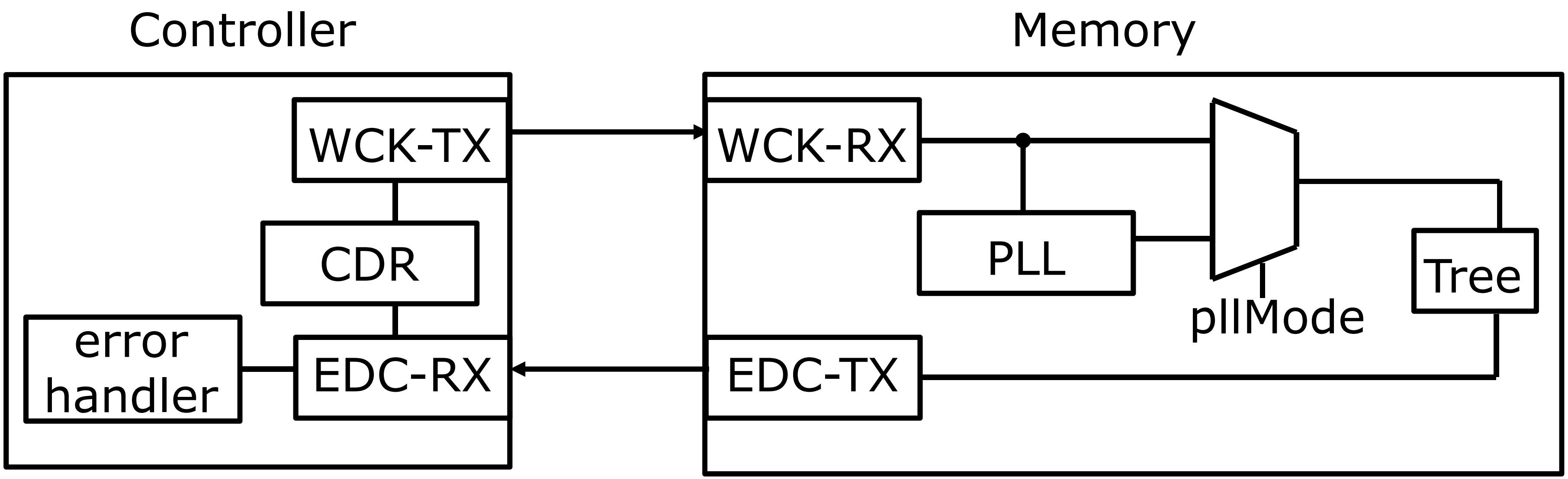
- Generalized implementation of TX-emphasis
  - Pre-emphasis: Boost transition
  - De-emphasis: Suppress non-transition

EDC Scheme



- Read-error => Controller initiates re-read
- Write-error => Controller initiates re-write
- Write-mask error => Cannot occur on slower ADD/CMD

Dual Use of EDC



- Dual functionality of EDC in GDDR5
- Back-channel for CRC-errors and echo-clock for CDR
  - Enables controller to track timing drifts (e.g. temperature)

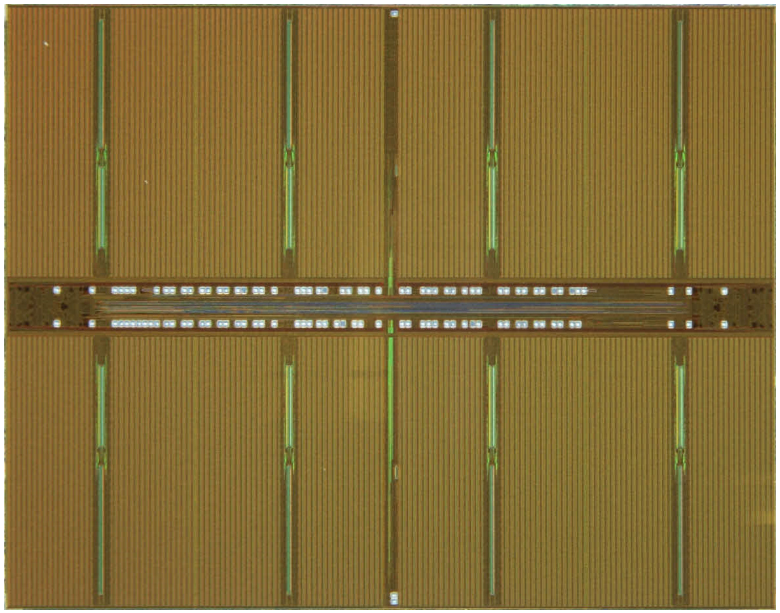
Outline

- Introduction
- DRAM Background
- DRAM Families
  - Main Memory DRAM
  - Graphics DRAM
  - LPDDR
  - TSV Solutions (HBM, HMC, Wide I/O)
- Summary and Conclusion

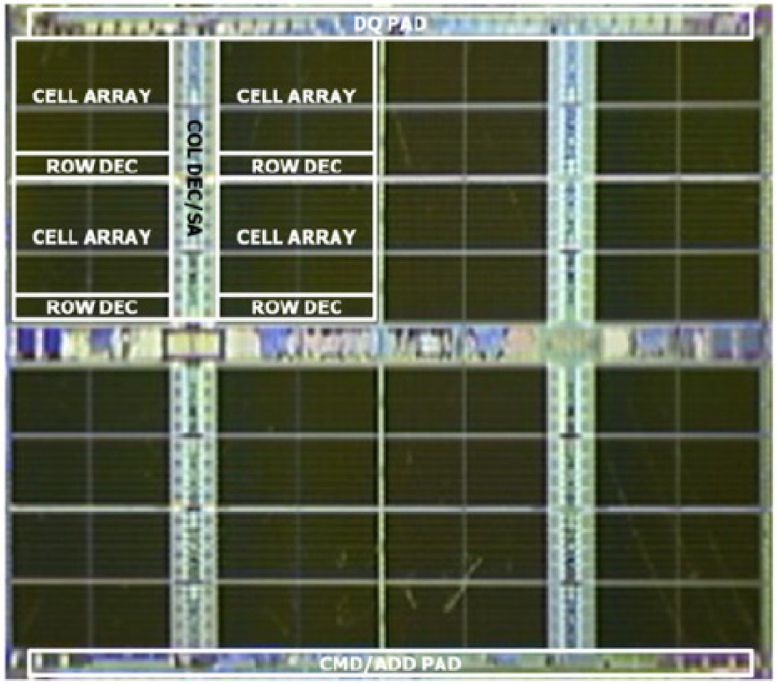
LPDDR

- Optimized for battery-sensitive applications
  - Reduce self-refresh current
  - Reduce leakage current
  - Reduce active current
- Minimized interface power
  - Avoid ODT as much as possible (LPDDR2/3)
  - Optimize for very small swing (LPDDR4)
- Minimized clocking power
  - Avoid using active clocking circuits (DLL/PLL)
  - Strobe-based interfaces
- Optimized for space-constraint applications

Edge Bond



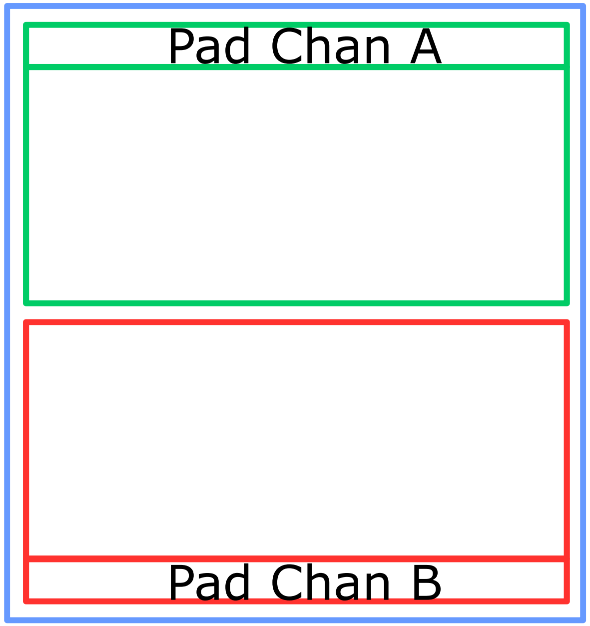
DDR3



LPDDR3
Y. C. Bae; ISSCC-2012

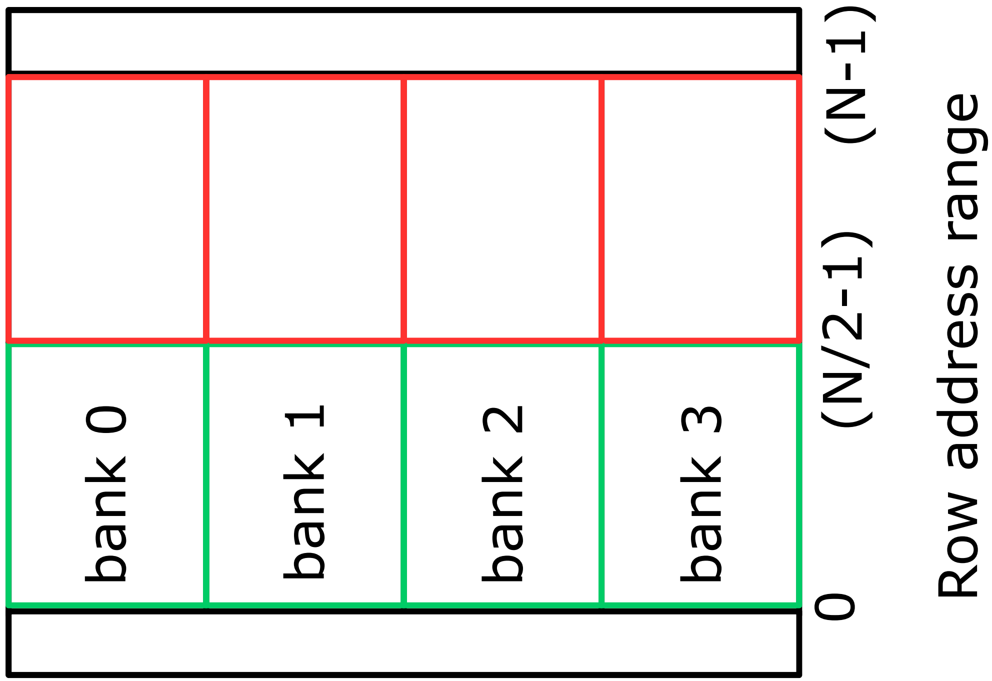
- Edge-bonding leads more easily to stacking

LPDDR4: Active Power Savings



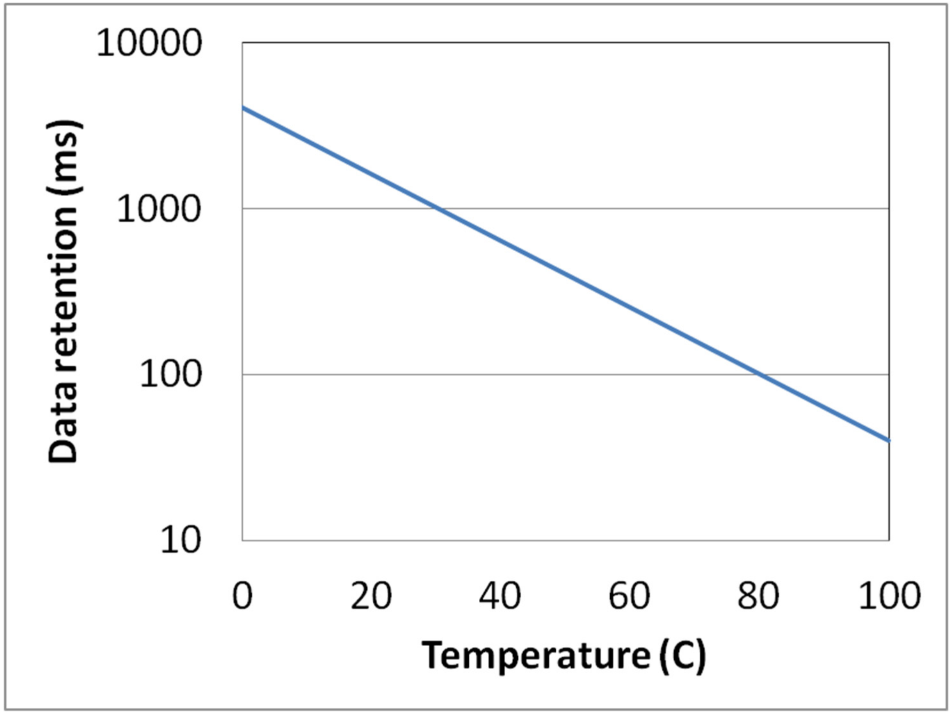
- LPDDR employs edge-pad bonding
- Active power reduction by independent channels
  - Routing length reduced

Limiting Self-Refresh Current (1)



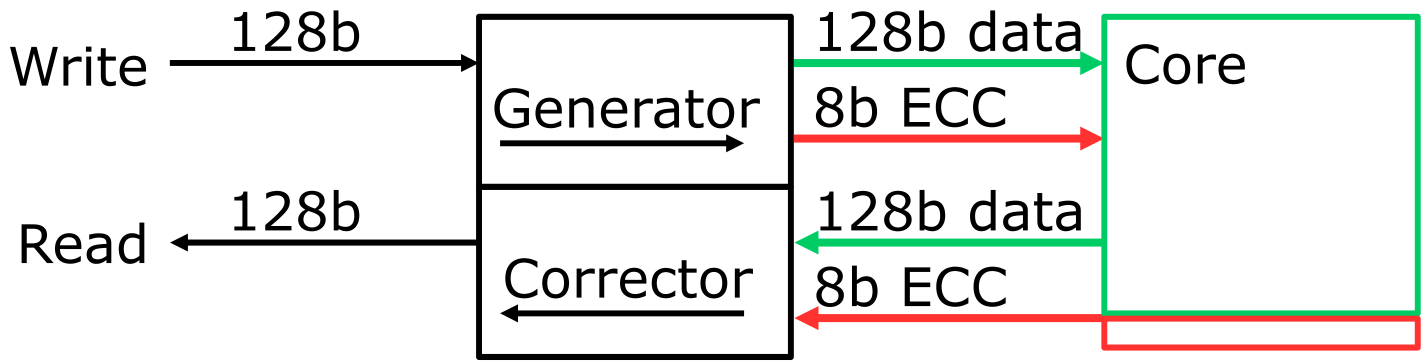
- For specific systems, refreshing full memory is not needed
  - Partial Array Self Refresh
- Use this fact by blocking self refresh to these regions
  - May block banks, rows
  - Example shows refresh of half the row-address range

Limiting Self-Refresh Current (2)



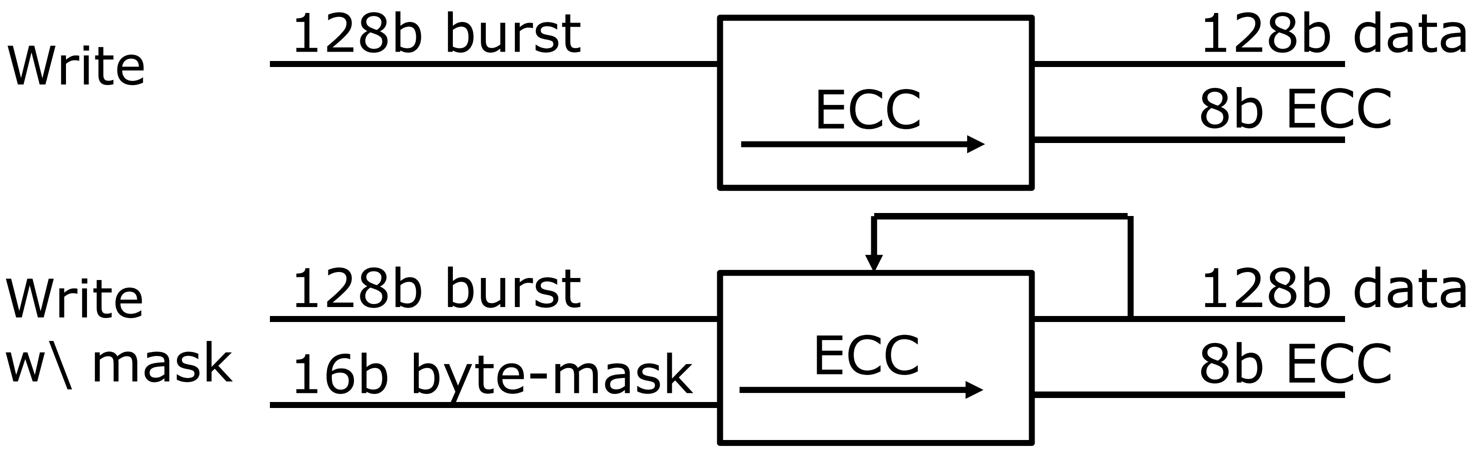
- DRAM retention improves with decreasing temperature
  - Rough guideline: retention doubles every 15°C
- Use on-die temperature to find optimized refresh rate
  - Temperature Controlled Self Refresh
  - Allow for programmable offset to account for hot-spots

On-Die ECC (1)



- On-die ECC is efficient to improve memory cell retention
- General approach
  - Calculate ECC-information while writing to core
  - Store ECC-information in additional memory cells
  - While reading, compare read-data to ECC-information
  - Use ECC to correct any potential errors

On-Die ECC (2)



- In a masked write, only partial data is written
- ECC calculation is possible by first reading back full burst
- LPDDR4 introduces special masked-write command
  - Timing sufficient for DRAM to perform hidden read

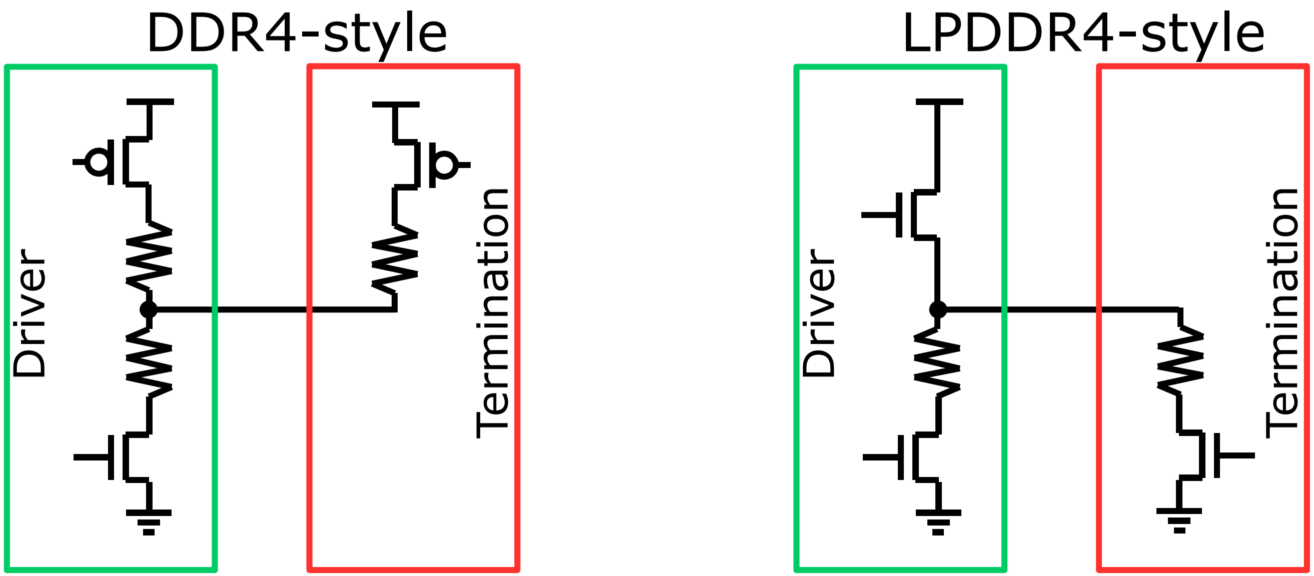
T.-Y. Oh; ISSCC-2014

Reducing Leakage Current

Standard methods

- Temperature-dependent body bias
- Power gate unused parts of periphery
  [e.g. B. H. Jong; ISSCC-2009]
- Power-gate unused parts of the core
  - Row and column drivers are typically main contributors

LPDDR4: Signaling



Y. C. Cho; VLSI 2013

- LPDDR4 introduces VSS-terminated high-speed interface
  - Termination is mandatory for 4Gbps operation
  - High-tap termination at low voltage swing is difficult
  - Ground acts as very defined reference plane
  - Pure nMOS driver can be implemented

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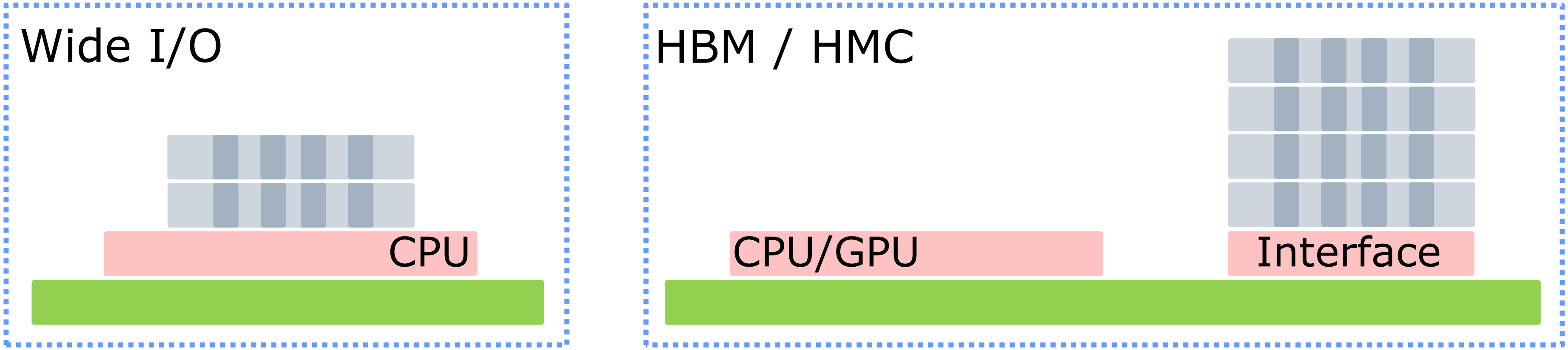
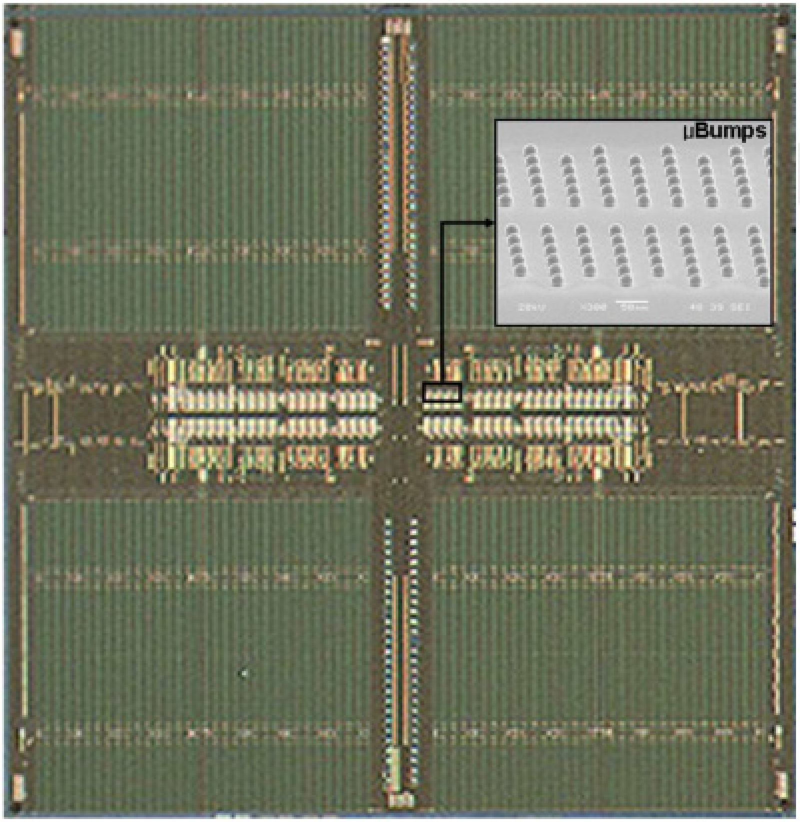
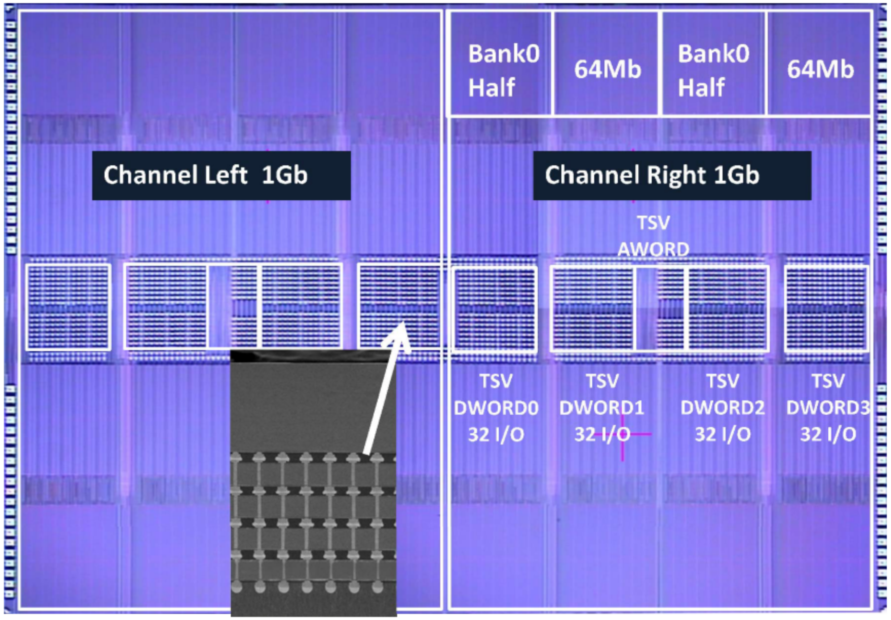
TSV/uBump based solutions



- Highly-parallel IO promises to increase system bandwidth
- Wide I/O solution for direct attachment to CPU
- HBM, HMC solutions for placement on GPU sub-system

TSV based devices





Wide I/O
J. S. Kim; ISSCC-2011

HBM
D. U. Lee; ISSCC-2014

Signaling

- uBumping allows very high ball-count
  - 100s to 1000s of connections
- Enables high overall bandwidth at moderate per-pin DR
  - Keep the IO as simple as possible
  - No termination
  - Low pin-capacitance
  - Strobe-based timing

Command processing

- Large number of interconnects enable complex commands
- Extend the channel concept of LPDDR4
  - Allow multiple independent channels
  - With concurrent row and column access
- HMC adds complex memory controller functionalities
- HMC adds serial link to host

Summary and Conclusion

	DDR3	DDR4	GDDR5	LPDDR4	WIO2	HBM
Application	Main	Main	Graphics	Mobile	Mobile	HPC
Voltage [V]	1.5	1.2	1.5	1.1	1.1	1.2
Data-rate [Gbps]	2	3	7	4	1	2
IO width	4, 8, 16	4, 8, 16	32	2x16	4x64 8x64	8x128
Data-rate [GB/s]	1, 2, 4	1.5, 3, 6	28	16	32 64	256
Bank group	W/Out	With	W/Out	W/Out	W/Out	With
tCCD [ns]	4	2.7	2.3	4	4	2
tRC [ns]	45	45	45	60	60	45
ODT	VDDQ/2	VDDQ	VDDQ	VSSQ	N/A	N/A
Package	BGA	BGA	BGA	MCP PoP	TSV uBump	TSV uBump

Summary and Conclusion

- Many building blocks are common between DRAM families
  - Strategies to hide the core
- Strong differences in the interfaces
  - Clocking, termination, IO-width, data-rate/pin
- Some differences vanish
  - LPDDR-techniques to reduce self-refresh current
  - GDDR-techniques to enhance data-rates
  - DDR4 adds TSV-stacking to increase density
- TSV and uBump promise a jump in the future

Papers to see at ISSCC 2015

- 10.3: "A 7.5mW 7.5Gb/s Mixed NRZ/Multi-Tone Serial-Data Transceiver for Multi-Drop Memory Interfaces in 40nm CMOS"
- 10.4: "A 5.8Gb/s Adaptive Integrating Duobinary-Based DFE Receiver for Multi-Drop Memory Interface"
- 10.6: "Continuous-Time Linear Equalization with Programmable Active-Peaking Transistor Arrays in a 14nm FinFET 2mW/Gb/s 16Gb/s 2-Tap Speculative DFE Receiver"
- 17.6: "1V 10Gb/s/pin Single-Ended Transceiver with Controllable Active-Inductor-Based Driver and Adaptively Calibrated Cascade-DFE for Post-LPDDR4 Interfaces"
- 17.7: "A Digital DLL with Hybrid DCC Using 2-Step Duty Error Extraction and 180° Phase Aligner for 2.67Gb/s/pin 16Gb 4-H Stack DDR4 SDRAM with TSVs"

References

- DRAM data-sheets from jedec.org
- K. Itoh, “VLSI Memory Chip Design”, Springer 2001
- B. Keeth et al., “DRAM Circuit Design”, IEEE 2000
- C. Kim et al., “High-Bandwidth Memory Interface”, Springer 2013

- S. J. Bae et al., “A 60nm 6Gb/s/pin GDDR5 Graphics DRAM with Multifaceted Clocking and ISI/SSN-Reduction Techniques”, ISSCC-2008
- K. Sohn et al., “A 1.2V 30nm 3.2Gb/s/pin 4Gb DDR4 SDRAM with Dual-Error Detection and PVT-Tolerant Data-Fetch Scheme”, ISSCC-2012, pp. 38-39
- Y. C. Cho et al., “A Sub-1.0V 20nm 5Gb/s/pin Post-LPDDR3 I/O interface with Low Voltage-Swing Terminated Logic and Adaptive Calibration Scheme For Mobile Application “, Symp. VLSI circuits 2013, pp. 240-241
- T. Na et al., “A Heterogeneous Dual DLL and Quantization Error Minimized ZQ Calibration for 30nm 1.2V 4Gb 3.2Gb/s/pin DDR4 SDRAM”, Symp. VLSI circuits 2013, pp. 242-243
- T. Y. Oh et al., “A 3.2Gb/s/pin 8Gb 1.0V LPDDR4 SDRAM with Integrated ECC Engine for Sub-1V DRAM Core Operation”, ISSCC-2014, pp. 430-431
- D. U. Lee et al., “A 1.2V 8Gb 8-Channel 128GB/s High-Bandwidth Memory (HBM) Stacked DRAM with Effective Microbump I/O Test Methods Using 29nm Process and TSV”, ISSCC-2014, pp. 432-433